MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE FEA

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification. [illegible] standards, of the issue listed in that issue of the Department of Defense Inde[illegible] f Specifications and Standards specified in the solicitation, form a part o[illegible] his specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.139 failures per million operating hours for class I or 0.473 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FEA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/295-1 for class I and M28787/295-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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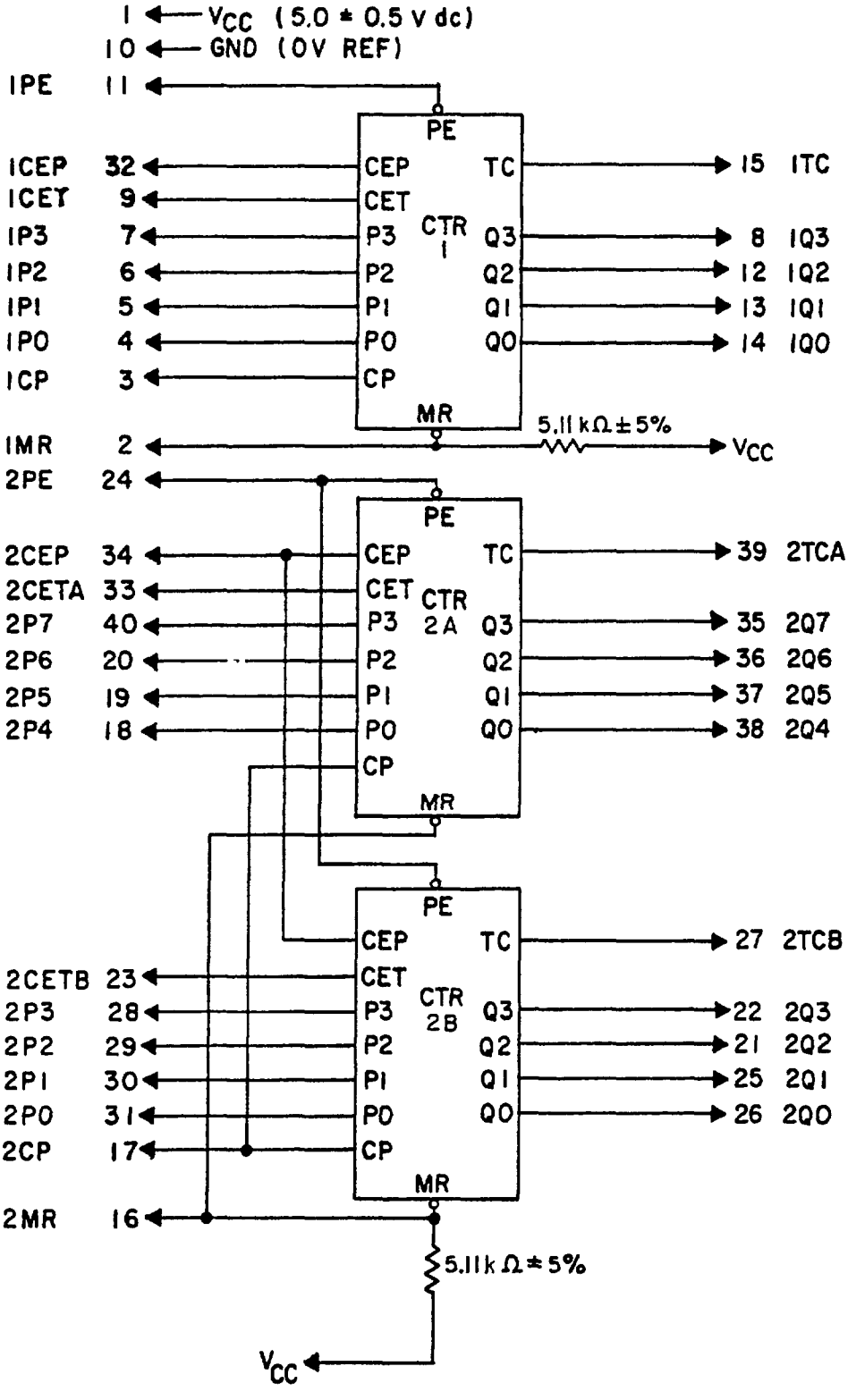


FIGURE 1. FEA module function diagram, binary counter.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B15	V_{OUT} (low-loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1,C3,C8, C11,C12, C27,C23	Input breakdown current	V	S	0	200	0	200	0	200	0	200	0	200	µA
C2,C4-C7, C9,C13-C16,C18-C21	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	µA
C10,C22	Input breakdown current	V	S	0	400	0	400	0	400	0	400	0	400	µA
D1-D15	Output short circuit current	VI	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
E1-E15	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1-F23	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G39	Functional	IX	X	← See table IX →										
H1-H15	Output voltage, low level	X	X	0	400	0	400	0	400	0	400	0	400	mV
J1-J15	Output volage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1,K3,K5	Input threshold, low level, check	XI	X	← See table XI →										
K2,K4,K6	Input threshold, high level, check	XI	X	← See table XI →										
L1,L3,L8, L11,L12, L17	Input current, low level	XII	X	-0.04	-0.87	-0.04	-0.87	-0.04	-0.87	-0.04	-0.87	-0.04	-0.87	mA
L2,L4-L7, L13-L16, L18-L21	Input current, low level	XII	X	-0.01	-0.43	-0.01	-0.43	-0.01	-0.43	-0.01	-0.43	-0.01	-0.43	mA
L9	Input current, low level	XII	X	-0.01	-1.51	-0.01	-1.51	-0.01	-1.51	-0.01	-1.51	-0.01	-1.51	mA
L10,L22	Input current, low level	XII	X	-0.08	-1.73	-0.08	-1.73	-0.08	-1.73	-0.08	-1.73	-0.08	-1.73	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L23	Input current, low level	XII	X	-0.08	-1.96	-0.08	-1.96	-0.08	-1.96	-0.08	-1.96	-0.08	-1.96	mA
M1,M3,M8, M11,M12, M17	Input current, high level	XIII	X	0	40	0	40	0	40	0	40	0	40	µA
M2,M4-M7, M13-M16, M18-M21	Input current, high level	XIII	X	0	20	0	20	0	20	0	20	0	20	µA
M9,M23	Input current, high level	XIII	X	-400	-610	-400	-610	-400	-610	-400	-610	-400	-610	µA
M10,M22	Input current, high level	XIII	X	0	80	0	80	0	80	0	80	0	80	µA
N	Power supply current, low level outputs	XIV	X	5	96	5	96	5	96	5	96	5	96	mA
P1-P15	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	50	5	50	5	55	5	50	5	58	ns
P16-P30	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	10	80	10	80	10	90	10	80	10	90	ns
Q1-Q15	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	18	3	18	3	20	3	18	3	21	ns
Q16-Q30	Transition time (t_{THL}), C_L = 150 pF	XV	Q	6	40	6	40	6	42	6	40	6	43	ns
R1,R6,R11	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	4	39	4	39	4	41	4	39	4	42	ns
R2-R5, R7-R10, R12-R15	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	3	28	3	28	3	30	3	28	3	31	ns
R16,R21, R26	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	5	49	5	49	5	51	5	49	5	52	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R17-R20, R22-R25, R27-R30	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4	38	4	38	4	40	4	38	4	41	ns
S1,S6,S11	Propgation delay time (t_{PHL}), C_L = 50 pF	XV	S	4	39	4	39	4	41	4	39	4	42	ns
S2,S5,S7-S7,S10, S12-S15	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	4	31	4	31	4	33	4	31	4	34	ns
S16,S21, S26	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	5	49	5	49	5	51	5	49	5	52	ns
S17-S20, S22-S25, S27-S30	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	41	4	41	4	43	4	41	4	44	ns
T1-T15	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	4	42	4	42	4	44	4	42	4	45	ns
T16-T30	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	5	52	5	52	5	54	5	52	5	55	ns
U1-U3	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	S	3	27	3	27	3	29	3	27	3	30	ns
U4-U6	Propagation delay time (t_{PLH}), C_L = 150 pF	XVII	Q	3	37	3	37	3	39	3	37	3	40	ns
V1-V3	Propagation delay time (t_{PHL}), C_L = 50 pF	XVII	S	3	27	3	27	3	29	3	27	3	30	ns
V4-V6	Propagation delay time (t_{PHL}), C_L = 150 pF	XVII	Q	3	37	3	37	3	39	3	37	3	40	ns
W1,W3,W5	Maximum clock frequency, minimum pulse width, PW1	XVIII	S	48	58	48	58	48	58	48	58	48	58	ns
W2,W4,W6	Maximum clock frequency, minimum pulse width, PW2	XVIII	S	38	48	38	48	38	48	38	48	38	48	ns
X1-X6	Setup time (t_S) check measuring t_C	XIX	S	3,100	3,300	3,100	3,300	3,100	3,300	3,100	3,300	3,100	3,300	ns
Y1-Y6	Hold time (t_H) check measuring t_C	XIX	S	3,100	3,300	3,100	3,300	3,100	3,300	3,100	3,300	3,100	3,300	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z3	Setup and hold time check, t_{PLH}	XX	S	3	29	3	29	3	31	3	29	3	33	ns
Z4-715	Setup and hold time check, low level, t_{PHL}	XX	S	3	31	3	31	3	33	3	31	3	35	ns
Z16-Z27	Setup and hold time check, high level, t_{PLH}	XX	S	3	28	3	28	3	30	3	28	3	32	ns
AA1-AA15	Output crosstalk, low level	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
BB1-BB12	Output crosstalk, high level	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
CC	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
DD	Power supply filter capacitance	XXIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
EE1-EE9, EE12-EE21	Input capacitance	XXIV	Q	2	25	2	25	2	25	2	25	2	25	pF
EE10,EE11, EE22,EE23	Input capacitance	XXIV	Q	4	40	4	40	4	40	4	40	4	40	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "X" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through DD shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition DD, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through EE shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	528	528	mW
Supply voltage	5 ±10 percent	5 ±10 percent	V dc
Input voltage			
Low level	-0.5 min, 0.7 max	-0.5 min, 0.7 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V

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	Class I	Class II	Unit
Input current, low level			
Inputs 2CEP, 1PE, 1CET, 2CETA, 2CETB, 1CP			
V_{IL} = 0.0 V	-0.87 max	-0.87 max	mA
V_{IL} = 0.4 V	-0.8 max	-0.8 max	mA
Input 2MR			
V_{IL} = 0.0 V	-1.96 max	-1.96 max	mA
V_{IL} = 0.4 V	-1.80 max	-1.80 max	mA
Inputs 2PE, 2CP			
V_{IL} = 0.0 V	-1.73 max	-1.73 max	mA
V_{IL} = 0.4 V	-1.6 max	-1.6 max	mA
Inputs 1CEP, 1PO-1P3, 2PO-2P7			
V_{IL} = 0.0 V	-0.43 max	-0.43 max	mA
V_{IL} = 0.4 V	-0.4 max	-0.4 max	mA
1MR			
V_{IL} = 0.0 V	-1.51 max	-1.51 max	mA
Input current, high level			
Inputs 2CEP, 1PE, 1CET, 2CETA, 2CETB, 1CP			
V_{IH} = 2.5 V	40 max	40 max	µA
V_{IH} = 7.0 V	200 max	200 max	µA
Inputs 2PE, 2CP			
V_{IH} = 2.5 V	80 max	80 max	µA
V_{IH} = 7.0 V	400 max	400 max	µA
Inputs 1CEP, 1PO-1P3, 2PO-2P7			
V_{IH} = 2.5 V	20 max	20 max	µA
V_{IH} = 7.0 V	100 max	100 max	µA
1MR, 2MR			
V_{IH} = 2.5 V	-610 max	-610 max	µA
Output voltage			
Low level (I_{OL} = 4.0 mA)	0.0 min, 0.4 max	0.0 min, 0.4 max	V
High level (I_{OH} = -400 µA)	2.5 min, V_{CC} max	2.5 min, V_{CC} max	V
Output drive current			
Low level (V_{OL} = 0.4 V)	4.0 max	4.0 max	mA
High level (V_{OH} = 2.5 V)	400 max	-400 max	µA
Clock input			
Frequency (C_L = 50 pF)	25 max	25 max	MHz
Pulse width	25 min	25 min	ns
Propagation delay time (t_{PHL})			
From clock to Q output			
C_L = 50 pF	4 min, 33 max	4 min, 34 max	ns
C_L = 150 pF	4 min, 43 max	4 min, 44 max	ns
From clock to TC output			
C_L = 50 pF	4 min, 41 max	4 min, 42 max	ns
C_L = 150 pF	5 min, 51 max	5 min, 52 max	ns
From CET to TC output			
C_L = 50 pF	3 min, 29 max	3 min, 30 max	ns
C_L = 150 pF	3 min, 39 max	3 min, 40 max	ns
From MR to Q output			
C_L = 50 pF	4 min, 44 max	4 min, 45 max	ns
C_L = 150 pF	5 min, 54 max	5 min, 55 max	ns
Propagation delay time (t_{PLH})			
From clock to Q output			
C_L = 50 pF	3 min, 30 max	3 min, 31 max	ns
C_L = 150 pF	4 min, 40 max	4 min, 41 max	ns
From clock to TC output			
C_L = 50 pF	4 min, 41 max	4 min, 42 max	ns
C_L = 150 pF	5 min, 51 max	5 min, 52 max	ns

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	Class I	Class II	Unit
From CET to TC output			
C_L = 50 pF	3 min, 29 max	3 min, 30 max	ns
C_L = 150 pF	3 min, 39 max	3 min, 40 max	ns
Data input setup time (t_S)			
High level	20 min	20 min	ns
Low level	20 min	20 min	ns
Data input hold time (t_H)			
High level	3 min	3 min	ns
Low level	3 min	3 min	ns
CET, CEP, PE input setup time (t_S)			
High level	20 min	20 min	ns
Low level	20 min	20 min	ns
CET, CEP, PE input hold time (t_H)			
High level	10 min	10 min	ns
Low level	10 min	10 min	ns
Reset pulse width	20 min	20 min	ns
Reset recovery time	20 min	20 min	ns

* 6.5 Module function. This module contains one completely independent 4-bit synchronous binary counter and two additional 4-bit synchronous binary counters that have common clock, reset, PE and CEP lines.

6.5.1 Input to output relationship. The synchronous binary circuit has the input to output relationship shown in table II. The FEA module contains three binary counters which are fully synchronous with the clock pulse. Each counter contains four master/slave flip-flops which are driven, in parallel, by the clock through a clock buffer. The master is inhibited from further change during the low-to-high transition of the clock. Data is transferred from the masters to the slaves and reflected at the outputs after the masters are inhibited. When the clock is high, the masters remain inhibited and the master/slave data path remains established. During the clock high-to-low transition, the slave is inhibited from further change, followed by the enabling of the masters for acceptance of new data from the counting logic or the parallel entry logic.

a. There are three control inputs: Parallel Enable (PE), Count Enable Parallel (CEP), and Count Enable Trickle (CET). These control inputs select the mode of operation as shown in table I. When the conditions for counting are satisfied, i.e., CEP, CET, and PE are high, the rising edge of the clock will change the counters to the state of the count sequence.

b. The counter can be synchronously preset from the four parallel inputs, (P0-P3) when PE is low. When the PE and clock are low, each master of the flip-flops is connected to the appropriate parallel input (P0-P3) and the slaves (outputs) are steady in their previous state. When the clock goes high, this information is transferred to the slaves and reflected at the outputs. The PE overrides both count enables presetting the counter low.

c. Terminal count is high when the counter is at a terminal count (state 15), and the CET is high.

d. The asynchronous master reset (MR) overrides all other inputs resetting each output low.

e. The PE, P0-P3, CET, and CEP inputs are completely edge triggered, i.e., responding only to the low-to-high transition of the clock line.

f. Long synchronous counters are constructed by connecting the TC output of the first counter to the CEP input of all other counters and the TC output of each counter after the first to the CET input of the next counter.

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6.5.2 TC (terminal count) = CET Q0 Q1 Q2 Q3. The TC output is asynchronous; therefore, the TC output should never be used outside the counting circuit as a synchronous output unless resynchronizing circuitry is added to compensate for the delay differences between the TC and Q outputs. When connected as a multistage counter (TC output connected to CET input), the delay t_{PHL} from CP to TC of a counter may exceed the delay t_{PLH} from CP to Q_0 of the following (adjacent) counter. Under the conditions of the first circuit changing from count 15_{10} to 0 and the following (MSB) circuit changing from count 14_{10} to 15_{10}, the TC output of that MSB counter could generate an illegal high level pulse.

6.5.3 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N392)

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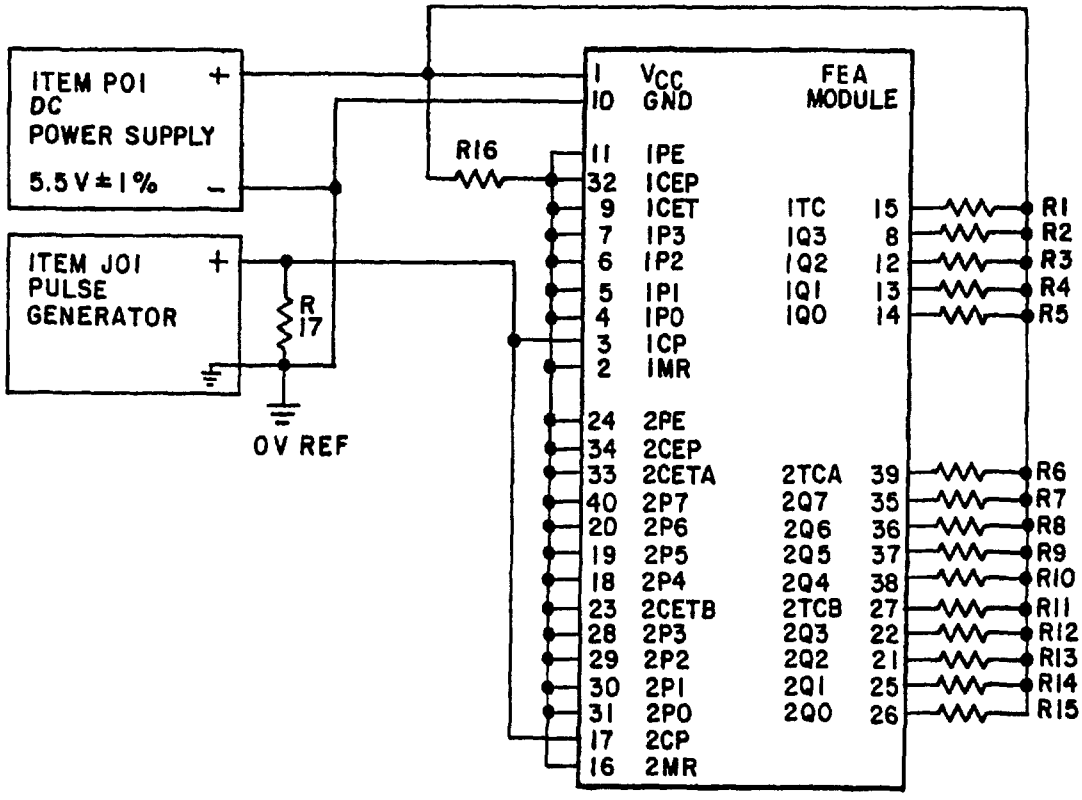
TABLE II. Input to output relationship. 1/-6/

Synchronous inputs before clock pulse							Required condition of asynchronous inputs	Outputs after clock pulse				
CEP	CET	PE	P3	P2	P1	P0	MR	TC	Q3	Q2	Q1	Q0
X	L	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
L	H	H	D3	D2	D1	D0	H	*	NC	NC	NC	NC
X	L	L	D3	D2	D1	D0	H	L	D3	D2	D1	D0
X	L	L	H	H	H	H	H	L	H	H	H	H
X	H	L	H	H	H	H	H	H	H	H	H	H
X	X	L	L	L	L	L	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	L
H	H	H	X	X	X	X	H	L	L	L	L	H
H	H	H	X	X	X	X	H	L	L	L	H	L
H	H	H	X	X	X	X	H	L	L	L	H	H
H	H	H	X	X	X	X	H	L	L	H	L	L
H	H	H	X	X	X	X	H	L	L	H	L	H
H	H	H	X	X	X	X	H	L	L	H	H	L
H	H	H	X	X	X	X	H	L	L	H	H	H
H	H	H	X	X	X	X	H	L	H	L	L	L
H	H	H	X	X	X	X	H	L	H	L	L	H
H	H	H	X	X	X	X	H	L	H	L	H	L
H	H	H	X	X	X	X	H	L	H	L	H	H
H	H	H	X	X	X	X	H	L	H	H	L	L
H	H	H	X	X	X	X	H	L	H	H	L	H
H	H	H	X	X	X	X	H	L	H	H	H	L
H	H	H	X	X	X	X	H	H	H	H	H	H
H	H	H	X	X	X	X	H	L	L	L	L	L

Asynchronous inputs		Required condition of synchronous inputs							Outputs				
MR	CET	CP	PE	CEP	P3	P2	P1	P0	TC	Q3	Q2	Q1	Q0
H	X	X	X	X	X	X	X	X	X	X	X	X	X
L	X	X	X	X	X	X	X	X	L	L	L	L	L
H	L	L	L	X	H	H	H	H	NC	NC	NC	NC	NC
H	L	H	L	X	H	H	H	H	NC	H	H	H	H
H	L	H	H	L	X	X	X	X	NC	NC	NC	NC	NC
H	L	H	H	H	X	X	X	X	NC	NC	NC	NC	NC
H	H	H	H	L	X	X	X	X	H	NC	NC	NC	NC
H	L	H	H	X	X	X	X	X	L	NC	NC	NC	NC
H	H	L	H	L	X	X	X	X	H	NC	NC	NC	NC
H	L	L	H	X	X	X	X	X	L	NC	NC	NC	NC

1/ NC = No change from previous state.
2/ D0-D3 = Known input and output data (either high level or low level).
3/ X = Unspecified inputs or outputs (either high level or low level).
4/ H = High-level.
5/ L = Low-level.
6/ * = If D0-D3 are high level, the TC is high level; otherwise TC is low level.

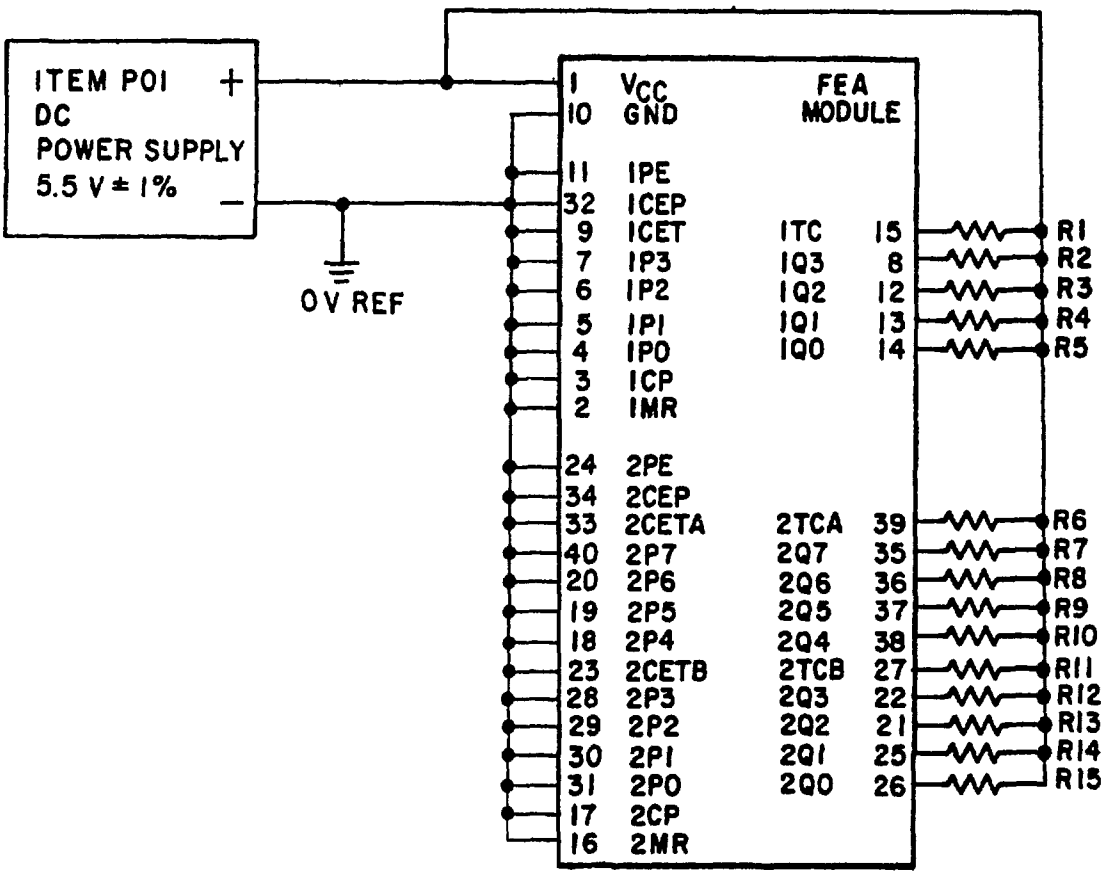
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NOTES:
1. R1-R15 = 1.27 kΩ ±1%, 1/4 W.
2. R16 = 1 kΩ ±5%, 1/4 W.
3. R17 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ±100 ns.
   PRR = 1 MHz ±20%.
   PA = 3.5 ±0.5 V.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 2. Operating temparature and life test circuit.

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NOTE: R1-R15 = 1.27 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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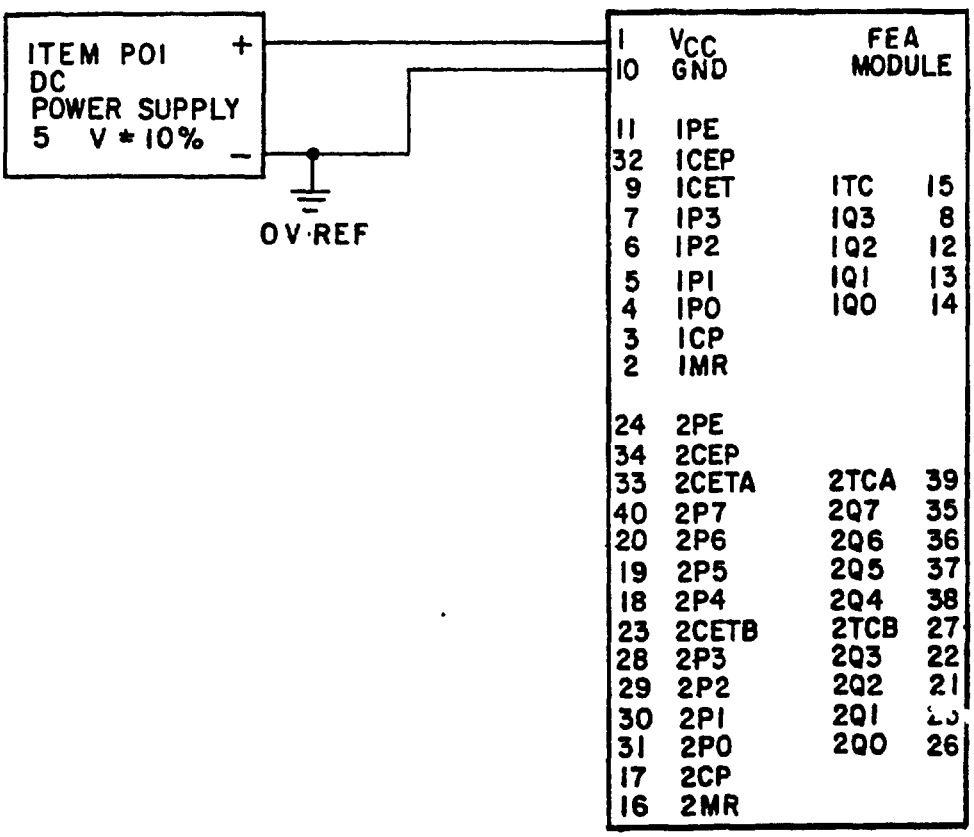


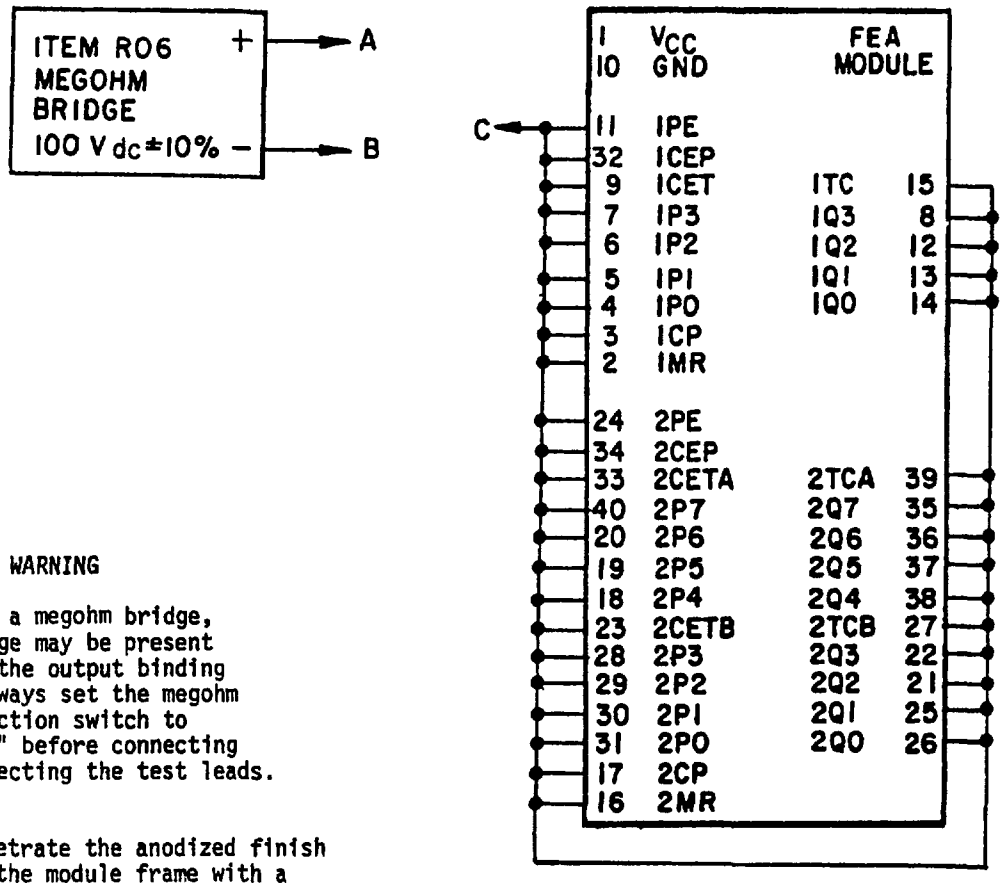
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

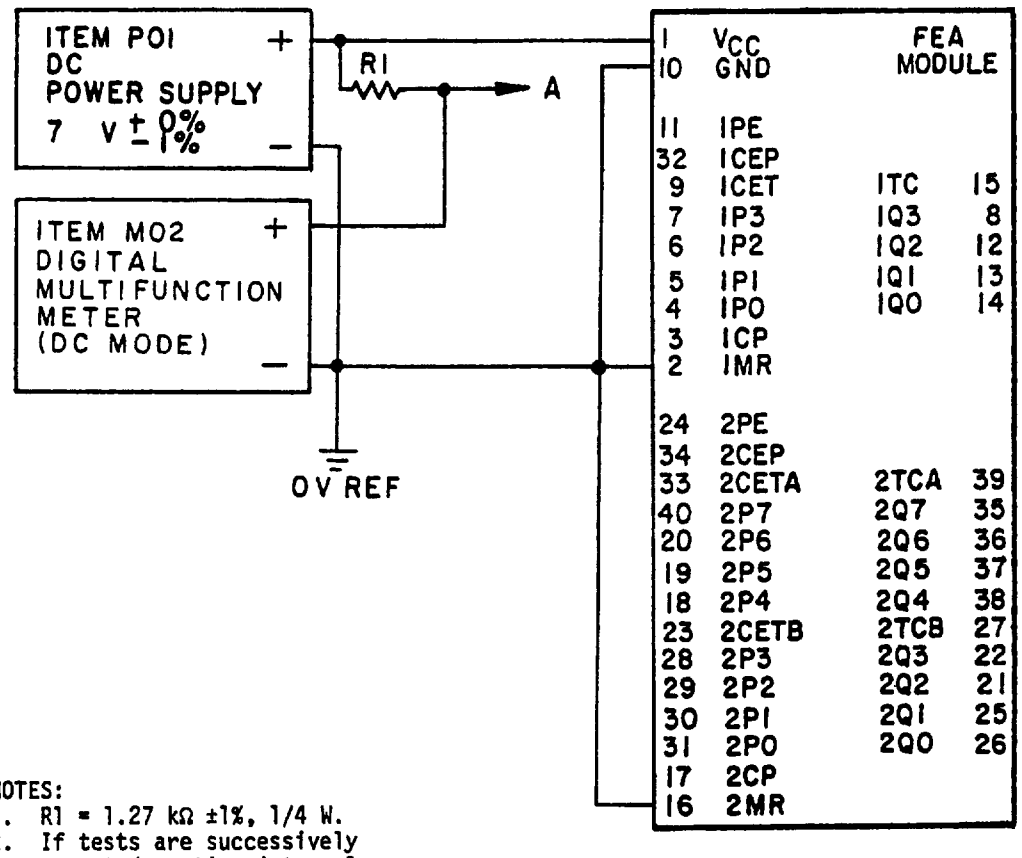
FIGURE 5. Isolation test circuit for table III.

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* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	1TC V_{OUT} (low loaded)	15	10	800	mV	6
B2	1Q3	8				
B3	1Q2	12				
B4	1Q1	13				
B5	1Q0	14				
B6	2TCA	39				
B7	2Q7	35				
B8	2Q6	36				
B9	2Q5	37				
B10	2Q4	38				
B11	2TCB	27				
B12	2Q3	22				
B13	2Q2	21				
B14	2Q1	25				
B15	2Q0	26				

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NOTES:
1. R1 = 1.27 kΩ ±1%, 1/4 W.
2. If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

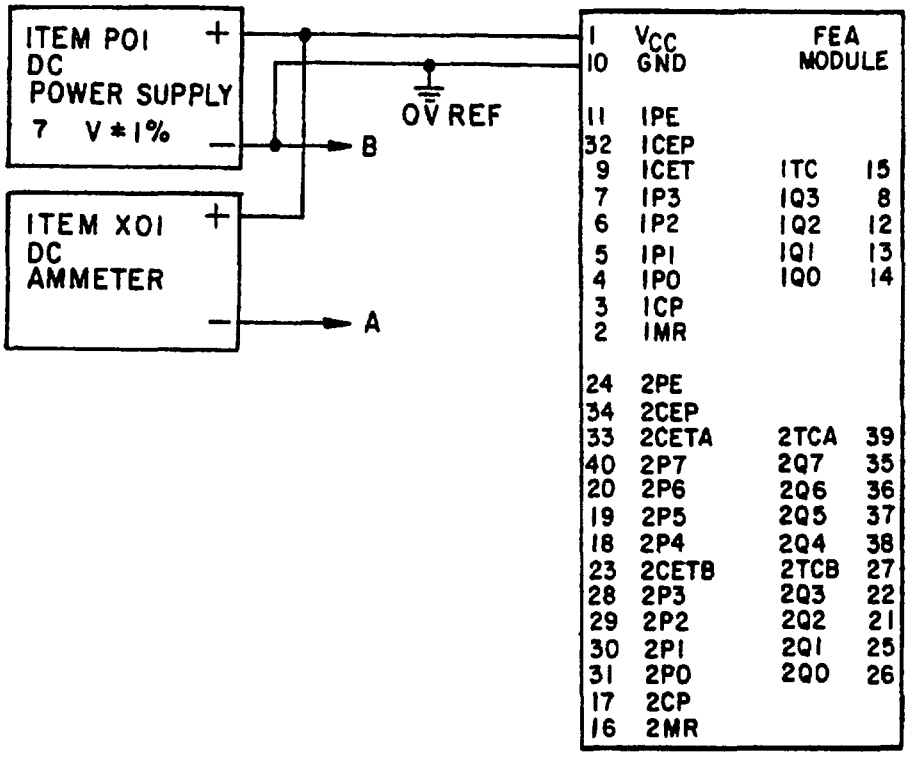
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1PE Input breakdown current	11	2,3,4,5,6,7,9,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40	0	200	µA	7
C2	1CEP	32	2,3,4,5,6,7,9,11,16,17,18, 19,20,23,24,28,29,30,31,33, 34,40		100		
C3	1CET	9	2,3,4,5,6,7,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40		200		
C4	1P3	7	2,3,4,5,6,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40		100		
C5	1P2	6	2,3,4,5,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40				
C6	1P1	5	2,3,4,6,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40				
C7	1PO	4	2,3,5,6,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40				
C8	1CP	3	2,4,5,6,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40		200		
C9	1MR	2	3,4,5,6,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 34,40		100		
C10	2PE	24	2,3,4,5,6,7,9,11,16,17,18,19, 20,23,28,29,30,31,32,33, 34,40		400		
C11	2CEP	34	2,3,4,5,6,7,9,11,16,17,18,19, 20,23,24,28,29,30,31,32,33, 40		200		

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C12	2CETA Input breakdown current	33	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,34,40	0	200	µA	7
C13	2P7	40	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,33,34		100		
C14	2P6	20	2,3,4,5,6,7,9,11,16,17,18,19,23,24,28,29,30,31,32,33,34,40				
C15	2P5	19	2,3,4,5,6,7,9,11,16,17,18,20,23,24,28,29,30,31,32,33,34,40				
C16	2P4	18	2,3,4,5,6,7,9,11,16,17,19,20,23,24,28,29,30,31,32,33,34,40				
C17	2CETB	23	2,3,4,5,6,7,9,11,16,17,18,19,20,24,28,29,30,31,32,33,34,40		200		
C18	2P3	28	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,29,30,31,32,33,34,40		100		
C19	2P2	29	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,30,31,32,33,34,40				
C20	2P1	30	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,31,32,33,34,40				
C21	2P0	31	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,32,33,34,40				
C22	2CP	17	2,3,4,5,6,7,9,11,16,18,19,20,23,24,28,29,30,31,32,33,34,40		400		
C23	2MR	16	2,3,4,5,6,7,9,11,17,18,19,20,23,24,28,29,30,31,32,33,34,40		200		

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

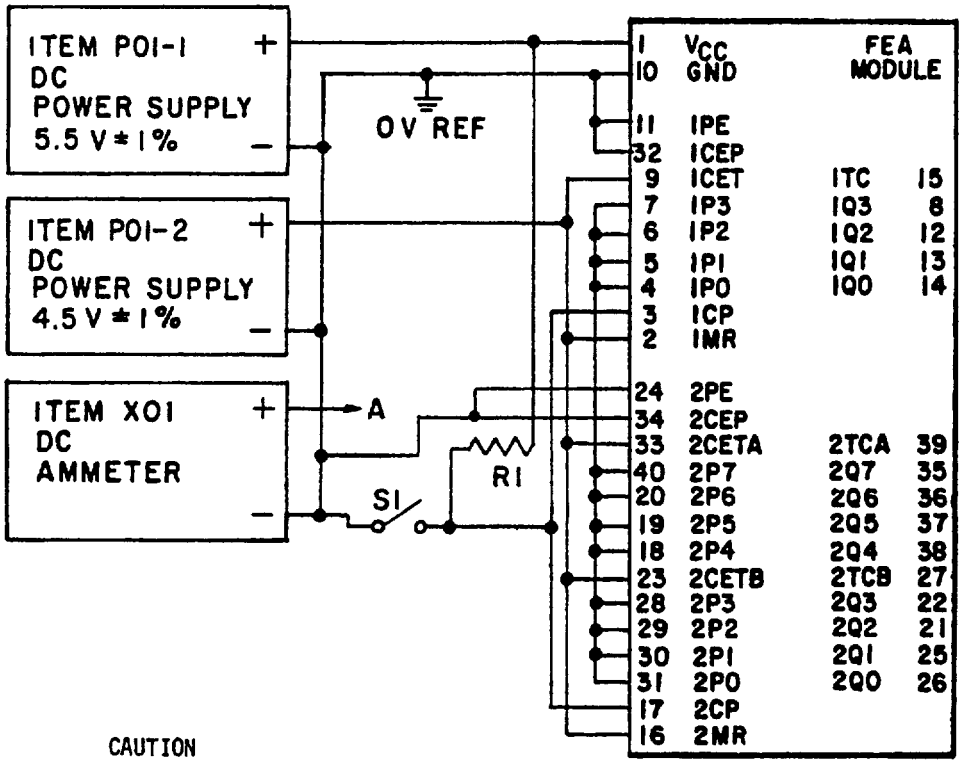
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect: A to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
D1	1TC Output short circuit current	15	-15	-100	mA	8
D2	1Q3	8				
D3	1Q2	12				
D4	1Q1	13				
D5	1Q0	14				
D6	2TCA	39				
D7	2Q7	35				
D8	2Q6	36				
D9	2Q5	37				
D10	2Q4	38				
D11	2TCB	27				
D12	2Q3	22				
D13	2Q2	21				
D14	2Q1	25				
D15	2Q0	26				

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. S1 = SPST switch.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.
4. The following sequence must be completed before performing test conditions D1-D15:

   a. With S1 open , connect item PO1-2 to pins 4, 5, 6, 7, 31, 30, 29, 28, 18, 19, 20, and 40.
   b. Close S1 and open.
   c. Connect pins 4, 5, 6, 7, 31, 30, 29, 28, 18, 19, 20, and 40 to OV REF and perform test conditions D1-D15 per table VI.

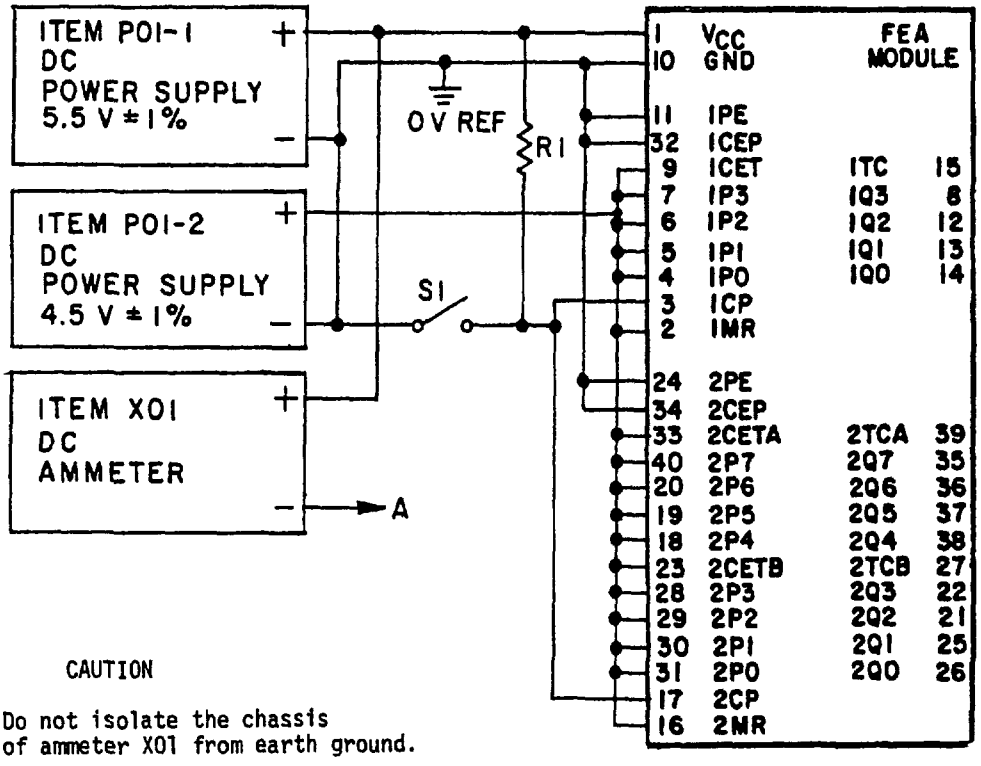
* FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1TC Output leakage current	15	0	1	mA	9
E2	1Q3	8				
E3	1Q2	12				
E4	1Q1	13				
E5	1Q0	14				
E6	2TCA	39				
E7	2Q7	35				
E8	2Q6	36				
E9	2Q5	37				
E10	2Q4	38				
E11	2TCB	27				
E12	2Q3	22				
E13	2Q2	21				
E14	2Q1	25				
E15	2Q0	26				

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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±5% 1/4 W.
2. S1 = SPST switch.
3. Before performing test conditions E1-E15, switch S1 must be closed, then opened.

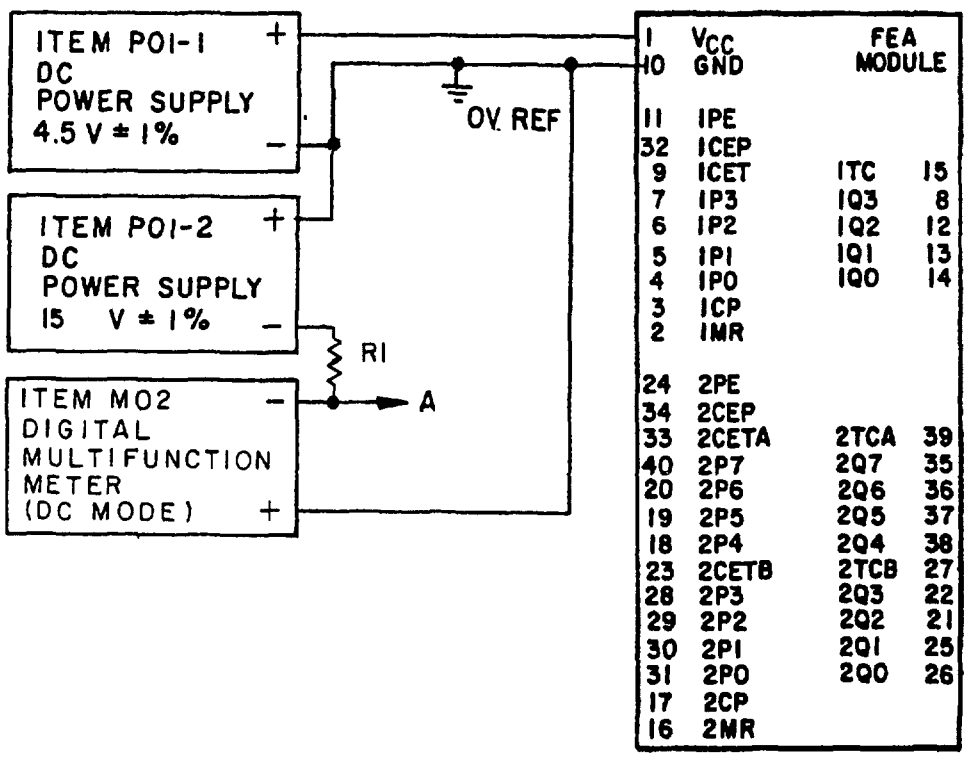
FIGURE 9. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1PE Input clamp diode voltage	11	-0.4	-1.5	V	10
F2	1CEP	32				
F3	1CET	9				
F4	1P3	7				
F5	1P2	6				
F6	1P1	5				
F7	1P0	4				
F8	1CP	3				
F9	1MR	2				
F10	2PE	24				
F11	2CEP	34				
F12	2CETA	33				
F13	2P7	40				
F14	2P6	20				
F15	2P5	19				
F16	2P4	18				
F17	2CETB	23				
F18	2P3	28				
F19	2P2	29				
F20	2P1	30				
F21	2P0	31				
F22	2CP	17				
F23	2MR	16				

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NOTE: R1 = 750Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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* TABLE IX. Functional test. 1/-3/

Test condition	Parameter	Connect			Output logic states															
		A to pin	B to pin	D to pin	1 Q 0	1 Q 1	1 Q 2	1 Q 3	1 T C	2 Q 4	2 Q 5	2 Q 6	2 Q 7	2 T C A	2 Q 0	2 Q 1	2 Q 2	2 Q 3	2 T C B	Test figure
G1	Functional	2,32,9,16,34, 33,23	4,5,6,7,11, 18,19,20,40, 24,31,30,29, 28	3,17	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11,12
G2		2,11,32,9,16, 24,34,33,23	4,5,6,7,18, 19,20,40,31, 30,29,28		H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	
G3					L	H	L	L	L	L	H	L	L	L	L	H	L	L	L	
G4					H	H	L	L	L	H	H	L	L	L	H	H	L	L	L	
G5					L	L	H	L	L	L	L	H	L	L	L	L	H	L	L	
G6					H	L	H	L	L	H	L	H	L	L	H	L	H	L	L	
G7					L	H	H	L	L	L	H	H	L	L	L	H	H	L	L	
G8					H	H	H	L	L	H	H	H	L	L	H	H	H	L	L	
G9					L	L	L	H	L	L	L	L	H	L	L	L	L	H	L	
G10					H	L	L	H	L	H	L	L	H	L	H	L	L	H	L	
G11					L	H	L	H	L	L	H	L	H	L	L	H	L	H	L	
G12					H	H	L	H	L	H	H	L	H	L	H	H	L	H	L	
G13					L	L	H	H	L	L	L	H	H	L	L	L	H	H	L	
G14					H	L	H	H	L	H	L	H	H	L	H	L	H	H	L	
G15					L	H	H	H	L	L	H	H	H	L	L	H	H	H	L	
G16					H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	
G17					L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	

See footnotes at end of table.

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* TABLE IX. Functional test - Continued. 1/-3/

Test condition	Parameter	Connect			Output logic states															Test figure
		A to pin	B to pin	D to pin	1 Q 0	1 Q 1	1 Q 2	1 Q 3	1 T C	2 Q 4	2 Q 5	2 Q 6	2 Q 7	2 T C A	2 Q 0	2 Q 1	2 Q 2	2 Q 3	2 T C B	
G18	Functional	4,5,6,7,11,32, 18,19,20,40, 24,34,31,30, 29,28	2,9,16,33,23	3,17	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	11,12
G19		4,5,6,7,2,11, 32,18,19,20, 40,16,24,34, 31,30,29,28	9,33,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G20		4,5,6,7,32,18, 19,20,40,34, 31,30,29,28	2,11,9,16, 24,33,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G21		4,5,6,7,2,32, 18,19,20,40, 16,34,31,30, 29,28	11,9,24,33,		H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	
G22		4,5,6,7,11,9, 18,19,20,40, 24,33,31,30, 29,28,23	2,32,16,34		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G23		4,5,6,7,2,11, 9,18,19,20,40, 16,24,33,31, 30,29,28,23	32,34		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G24		5,2,32,9,19, 16,34,33,30,23	4,6,7,11,18, 20,40,24,31, 29,28		L	H	L	L	L	L	H	L	L	L	L	H	L	L	L	
G25		2,32,9,16,34, 33,23	4,5,6,7,11, 18,19,20,40, 24,31,30,29, 28		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G26		6,2,32,9,20, 16,34,33,29,23	4,5,7,11,18, 19,40,24,31, 30,28		L	L	H	L	L	L	L	H	L	L	L	L	H	L	L	
G27		2,32,9,16,34, 33,23	4,5,6,7,11, 18,19,20,40, 24,31,30,29, 28		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	

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* TABLE IX. Functional test - Continued. 1/-3/

Test condition	Parameter	Connect			Output logic states														Test figure
		A to pin	B to pin	D to pin	1 Q 0	1 Q 1	1 Q 2	1 Q 3	1 T C	2 Q 4	2 Q 5	2 Q 6	2 Q 7	2 T C A	2 Q 0	2 Q 1	2 Q 2	2 Q 3	2 T C B	
G28	Functional	7,2,32,9,40, 16,34,33,28, 23	4,5,6,11,18, 19,20,24,31, 30,29	3,17	L	L	L	H	L	L	L	L	H	L	L	L	L	H	L	11,12
G29		2,32,9,16,34, 33,23	4,5,6,7,11, 18,19,20,40, 24,31,30,29, 28		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G30		4,2,9,18,16, 33,31,23	5,6,7,11,32, 19,20,40,24, 34,30,29,28		H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	
G31		4,2,11,9,18, 16,24,33,31,23	5,6,7,32,19, 20,40,34,30, 29,28		H	L	L	L	L	H	L	L	L	L	H	L	L	L	L	
G32		2,16	4,5,6,7,11, 32,9,18,19, 20,40,24,34, 33,31,30,29, 28,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G33		4,5,2,18,19, 16,31,30	6,7,11,32,9, 20,40,24,34, 33,29,28,23		H	H	L	L	L	H	H	L	L	L	H	H	L	L	L	
G34		2,16	4,5,6,7,11, 32,9,18,19, 20,40,24,34, 33,31,30,29, 28,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	
G35		4,5,6,2,18,19, 20,16,31,30,29	7,11,32,9, 40,24,34,33, 28,23		H	H	H	L	L	H	H	H	L	L	H	H	H	L	L	
G36		2,16	4,5,6,7,11, 32,9,18,19, 20,40,24,34, 33,31,30,29, 28,23		L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	

See footnotes at end of table.

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* TABLE IX. Functional test - Continued. 1/-3/

Test condition	Parameter	Connect			Output logic states															Test figure
		A to pin	B to pin	D to pin	1Q0	1Q1	1Q2	1Q3	1TC	2Q4	2Q5	2Q6	2Q7	2TCA	2Q0	2Q1	2Q2	2Q3	2TCB	
G37	Functional	4,5,6,7,2,18, 19,20,40,16, 31,30,29,28	11,32,9,24, 34,33,23	3,17	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	11,12
G38	Functional	4,5,6,7,2,11, 18,19,20,40, 16,24,31,30, 29,28	32,9,34,33, 23	3,17	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	11,12
G39	Functional	4,5,6,7,2,11, 18,19,20,40, 16,24,31,30, 29,28	32,9,34,33, 23	3,17	H	H	H	H	L	H	H	H	H	L	H	H	H	H	L	11,12

1/ Output logic states:

L = Low-voltage level ≤0.4 V.
H = High-voltage level ≥2.5 V.

2/ For test conditions G1-G39 verification of the output logic states in footnote 1 is obtained by connecting item MO2, point C, to each output in succession after input conditions have been accomplished. Set item JO1 to the single pulse mode and pulse once before each set of readings. Test conditions G1-G39 are to be performed in sequence.

3/ 1QO, 1Q1, 1Q2, 1Q3, 1TC, 2Q4, 2Q5, 2Q6, 2Q7, 2TCA, 2QO, 2Q1, 2Q2, 2Q3, and 2TCB output logic states are measured at output pins 14, 13, 12, 8, 15, 38, 37, 36, 35, 39, 26, 25, 21, 22, and 27, respectively.

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* TABLE X. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
H1	1TC	Output voltage, low level	4,5,6,7, 32	9,11,E	15	3	8	2	0	400	mV	11,12
H2	1Q3		4,5,6,32, 9	E,7,11	8							
H3	1Q2		4,5,7,32, 9	E,6,11	12							
H4	1Q1		4,6,7,32, 9	E,5,11	13							
H5	1Q0		5,6,7,32, 9	E,4,11	14							
H6	2TCA		18,19,20, 40,34	24,33,E	39	17		16				
H7	2Q7		18,19,20, 33,34	24,40,E	35							
H8	2Q6		18,19,40, 33,34	24,20,E	36							
H9	2Q5		18,20,40, 33,34	24,19,E	37							
H10	2Q4		19,20,40, 33,34	24,18,E	38							
H11	2TCB		28,29,30, 31,34	24,23,E	27							
H12	2Q3		29,30,34, 23,31	24,28,E	22							
H13	2Q2		28,30,34, 31,23	24,29,E	21							
H14	2Q1		28,29,31, 34,23	24,30,E	25							
H15	2Q0		28,29,30, 34,23	24,31,E	26							

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* TABLE X. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J1	1TC Output voltage, high level	4,5,6,7, 9,32	11,E	15	3	B	2	2.5	4.5 V	V	11,12
J2	1Q3	7	4,5,6, 9,32, 11,E	8							
J3	1Q2	6	4,5,7, 9,32, 11,E	12							
J4	1Q1	5	4,6,7, 9,32, 11,E	13							
J5	1Q0	4	5,6,7, 9,32, 11,E	14							
J6	2TCA	18,19,20, 40,33,34	24,E	39	17		16				
J7	2Q7	40	18,19, 20,33, 34,24,E	35							
J8	2Q6	20	18,19, 40,33, 34,24, E	36							
J9	2Q5	19	18,20, 40,33, 34,24,E	37							
J10	2Q4	18	19,20, 40,33, 34,24, E	38							
J11	2TCB	28,29,30, 31,23,34	24,E	27							

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* TABLE X. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
J12	2Q3	Output voltage, high level	28	29,30, 31,23, 24,34,E	22	17	8	16	2.5	4.5	V	11,12
J13	2Q2	↓	29	28,30, 31,23, 24,34,E	21	↓	↓	↓	↓	↓	↓	↓
J14	2Q1	↓	30	28,29, 31,23, 24,34, E	25	↓	↓	↓	↓	↓	↓	↓
J15	2Q0	↓	31	28,29, 30,23, 24,34,E	26	↓	↓	↓	↓	↓	↓	↓

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* TABLE XI. Input threshold (low-level and high-level) check test. 1/

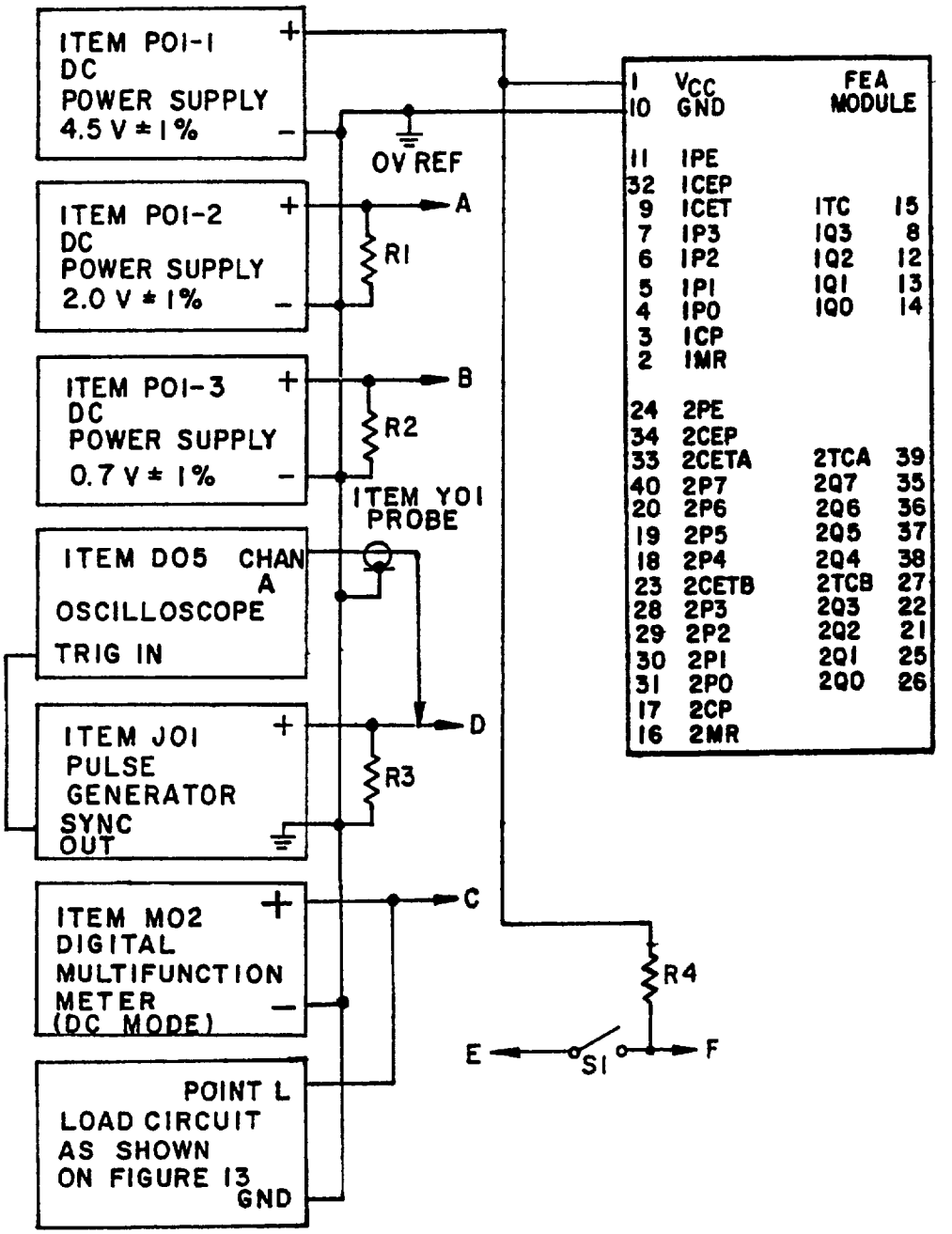
Test condition	Parameter		Connect						Output logic states					Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	1 T C	1 Q 3	1 Q 2	1 Q 1	1 Q 0	
K1	1MR	Input threshold, low level, check	4,5,6,7, 9,32	11,E	Open	2	B	3	L	L	L	L	L	11,12
K2	1CET,1CEP,1PE	Input threshold, high level, check	9,11,32	7,E,4, 5,6	Open	3	B	2	L	L	L	L	L	11,12
									2 T C A	2 Q 7	2 Q 6	2 Q 5	2 Q 4	
K3	2MR	Input threshold, low level, check	18,19,20, 40,33,34	24,E	Open	16	B	17	L	L	L	L	L	11,12
K4	2CETA,2CEP,2PE	Input threshold, high level, check	33,24,34	18,19, 20,40,E	Open	17	B	16	L	L	L	L	L	11,12
									2 T C B	2 Q 3	2 Q 2	2 Q 1	2 Q 0	
K5	2MR	Input threshold, low level, check	23,28,29, 30,31,34	24,E	Open	16	B	17	L	L	L	L	L	11,12
K6	2CETB,2CEP,2PE	Input threshold, high level, check	23,24,34	28,29, 30,31,E	Open	17	B	16	L	L	L	L	H	11,12

1/ Output logic states:

L = Low level = 0.0 to 0.4 V.

H = High level = 2.5 to 4.5 V.

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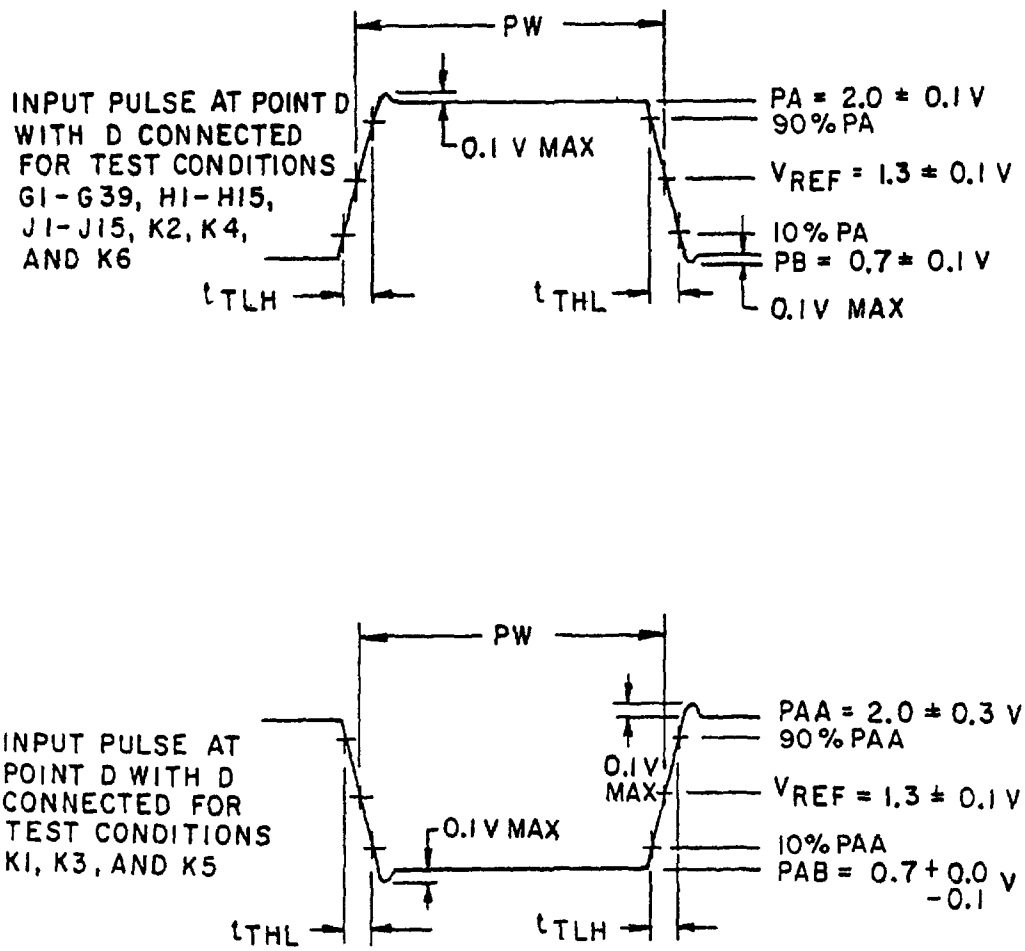
* FIGURE 11. Functional, output voltage (low-level and high-level), and input threshold (low-level and high-level) check test circuit for tables IX, X, and XI.

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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1/2 W.
2. R4 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. Test conditions G1-G39:
   Use item M02 to measure output logic states: L ≤ 0.4 V, and H ≥ 2.5 V. Item J01 is in single pulse mode, pulse once before each set of readings.
5. Test conditions H1-H15, J1-J15, K2, K4, and K6:
   After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The voltage levels and output logic states shall be as specified in tables X and XI.
6. Test conditions K1, K3 and K5:
   Initially switch S1 shall be closed. After making all circuit connections for a test condition, open switch S1 for a minimum of 100 ns, then close switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table XI.

* FIGURE 11. Functional, output voltage (low-level and high-level), and input threshold (low-level and high-level) check test circuit for tables IX, X, and XI - Continued.

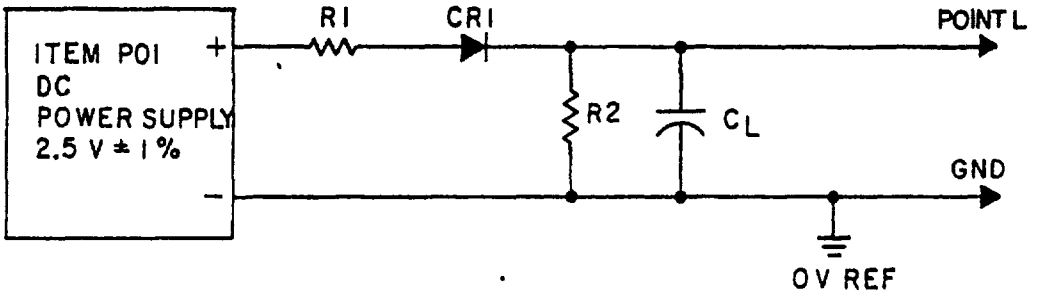
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NOTES:
1. PW = 500 ±100 ns.
2. $t_{TLH} = t_{THL} = 30 \pm 5$ ns.

* FIGURE 12. Functional, output voltage (low-level and high-level), and input threshold (low-level and high-level) check test circuit waveforms for tables IX, X, and XI.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 4.0 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions G1-G39, H1-H15, J1-J15 and K1-K6.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

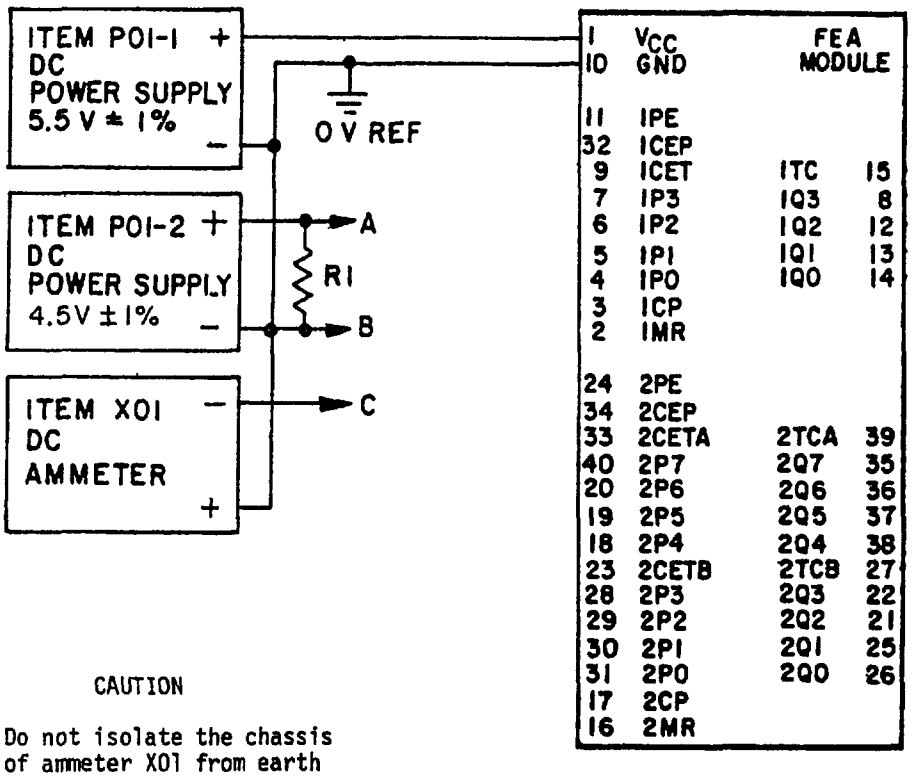
* FIGURE 13. FEA logic load circuit.

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TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1PE Input current, low level	9, 32	Open	11	-0.04	-0.87	mA	14
L2	1CEP	9, 11	Open	32	-0.01	-0.43		
L3	1CET	11, 32	Open	9	-0.04	-0.87		
L4	1P3	Open	11	7	-0.01	-0.43		
L5	1P2			6				
L6	1P1			5				
L7	1PO			4				
L8	1CP		Open	3	-0.04	-0.87		
L9	1MR			2	-0.01	-1.51		
L10	2PE	23, 33, 34		24	-0.08	-1.73		
L11	2CEP	23, 24, 33		34	-0.04	-0.87		
L12	2CETA	24, 34		33	-0.04	-0.87		
L13	2P7	Open	24	40	-0.01	-0.43		
L14	2P6			20				
L15	2P5			19				
L16	2P4			18				
L17	2CETB	24, 34	Open	23	-0.04	-0.87		
L18	2P3	Open	24	28	-0.01	-0.43		
L19	2P2			29				
L20	2P1			30				
L21	2PO			31				
L22	2CP		Open	17	-0.08	-1.73		
L23	2MR		Open	16	-0.08	-1.96		

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 14. Input current (low-level) test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1PE Input current, high level	Open	2,3,4,5,6,7,9,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	11	0	40	µA	15
M2	1CEP	Open	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,33,34,40	32		20		
M3	1CET	Open	2,3,4,5,6,7,11,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	9		40		
M4	1P3	11	2,3,4,5,6,9,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	7		20		
M5	1P2		2,3,4,5,7,9,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	6				
M6	1P1		2,3,4,6,7,9,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	5				
M7	1PO		2,3,5,6,7,9,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	4				
M8	1CP	Open	2,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	3		40		
M9	1MR	Open	3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,33,34,40	2	-400	-610		
M10	2PE	Open	2,3,4,5,6,7,9,11,16,17,18,19,20,23,28,29,30,31,32,33,34,40	24	0	80		

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M11	2CEP Input current, high level	Open	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,33,40	34	0	40	µA	15
M12	2CETA	Open	2,3,4,5,6,7,9,11,16,17,18,19,20,23,24,28,29,30,31,32,34,40	33		40		
M13	2P7	24	2,3,4,5,6,7,9,11,16,17,18,19,20,23,28,29,30,31,32,33,34	40		20		
M14	2P6		2,3,4,5,6,7,9,11,16,17,18,19,23,28,29,30,31,32,33,34,40	20				
M15	2P5		2,3,4,5,6,7,9,11,16,17,18,20,23,28,29,30,31,32,33,34,40	19				
M16	2P4		2,3,4,5,6,7,9,11,16,17,19,20,23,28,29,30,31,32,33,34,40	18				
M17	2CETB	Open	2,3,4,5,6,7,9,11,16,17,18,19,20,24,28,29,30,31,32,33,34,40	23		40		
M18	2P3	24	2,3,4,5,6,7,9,11,16,17,18,19,20,23,29,30,31,32,33,34,40	28		20		
M19	2P2	24	2,3,4,5,6,7,9,11,16,17,18,19,20,23,28,30,31,32,33,34,40	29		20		
M20	2P1	24	2,3,4,5,6,7,9,11,16,17,18,19,20,23,28,29,31,32,33,34,40	30		20		

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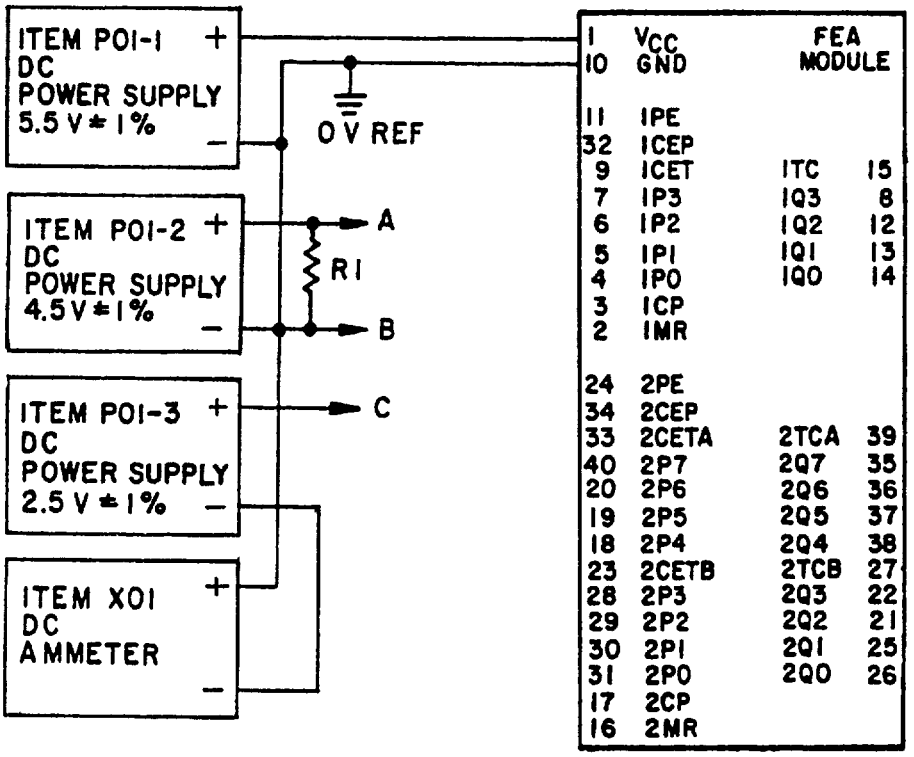
TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M21	2PO Input current, high level	24	2,3,4,5,6,7,9,11,16, 17,18,19,20,23,28,29, 30,32,33,34,40	31	0	20	µA	15
M22	2CP Input current, high level	Open	2,3,4,5,6,7,9,11,16, 18,19,20,23,24,28,29, 30,31,32,33,34,40	17	0	80	µA	15
M23	2MR Input current, high level	Open	2,3,4,5,6,7,9,11,17, 18,19,20,23,24,28,29, 30,31,32,33,34,40	16	-400	-610	µA	15

TABLE XIV. Power supply current (low-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N	Power supply current, low level outputs	2,16	3,4,5,6,7,9,11,17,18,19,20, 23,24,28,29,30,31,32,33,34, 40	5	96	mA	16

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 15. Input current (high-level) test circuit for table XIII.

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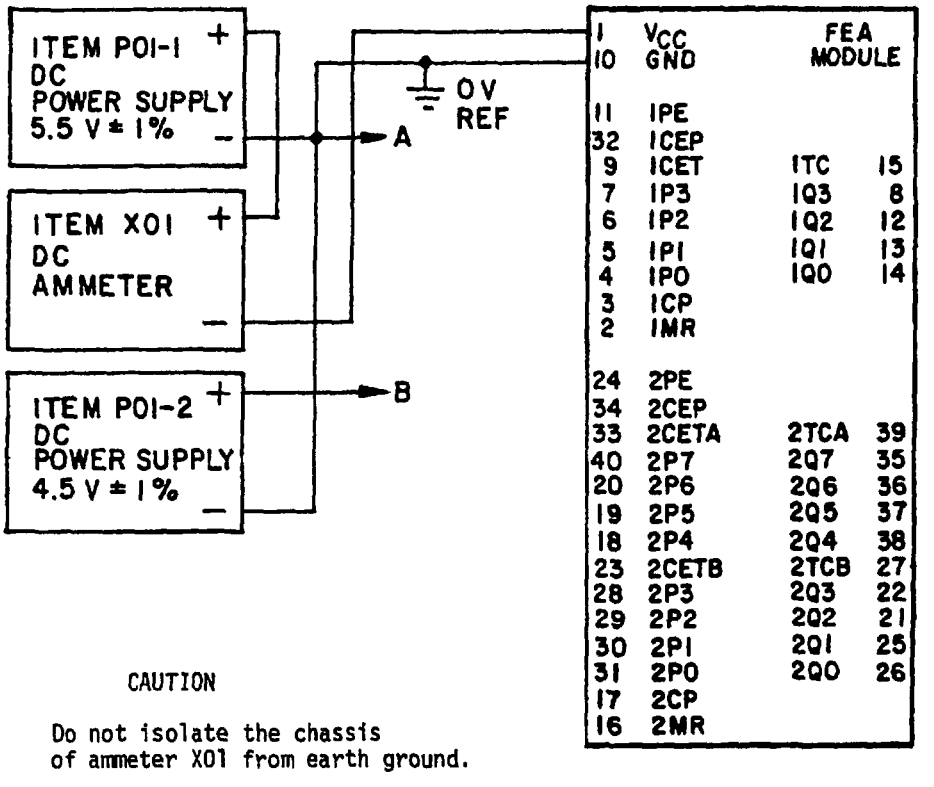


FIGURE 16. Power supply current (low-level outputs) test circuit for table XIV.

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
P1	1TC Transition time (t_{TLH}), C_L = 50 pF	15	5	50	ns	17,18
P2	1Q3	8				
P3	1Q2	12				
P4	1Q1	13				
P5	1Q0	14				
P6	2TCA	39				
P7	2Q7	35				
P8	2Q6	36				
P9	2Q5	37				
P10	2Q4	38				
P11	2TCB	27				
P12	2Q3	22				
P13	2Q2	21				
P14	2Q1	25				
P15	2Q0	26				
P16	1TC Transition time (t_{TLH}), C_L = 150 pF	15	10	80		
P17	1Q3	8				
P18	1Q2	12				
P19	1Q1	13				
P20	1Q0	14				

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test - Continued.

Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
P21	2TCA	Transition time (t_{TLH}), C_L = 150 pF	39	10	80	ns	17,18
P22	2Q7		35				
P23	2Q6		36				
P24	2Q5		37				
P25	2Q4		38				
P26	2TCB		27				
P27	2Q3		22				
P28	2Q2		21				
P29	2Q1		25				
P30	2Q0		26				
Q1	1TC	Transition time (t_{THL}), C_L = 50 pF	15	3	18		
Q2	1Q3		8				
Q3	1Q2		12				
Q4	1Q1		13				
Q5	1Q0		14				
Q6	2TCA		39				
Q7	2Q7		35				
Q8	2Q6		36				
Q9	2Q5		37				
Q10	2Q4		38				

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q11	2TCB	Transition time (t_{THL}), C_L = 50 pF	27	3	18	ns	17,18
Q12	2Q3		22				
Q13	2Q2		21				
Q14	2Q1		25				
Q15	2Q0		26				
Q16	1TC	Transition time (t_{THL}), C_L = 150 pF	15	6	40		
Q17	1Q3		8				
Q18	1Q2		12				
Q19	1Q1		13				
Q20	1Q0		14				
Q21	2TCA		39				
Q22	2Q7		35				
Q23	2Q6		36				
Q24	2Q5		37				
Q25	2Q4		38				
Q26	2TCB		27				
Q27	2Q3		22				
Q28	2Q2		21				
Q29	2Q1		25				
Q30	2Q0		26				

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R1	CP to 1TC	Propagation delay time (t_{PLH}), C_L = 50 pF	15	4	39	ns	17,18
R2	CP to 1Q3		8	3	28		
R3	CP to 1Q2		12				
R4	CP to 1Q1		13				
R5	CP to 1Q0		14				
R6	CP to 2TCA		39	4	39		
R7	CP to 2Q7		35	3	28		
R8	CP to 2Q6		36				
R9	CP to 2Q5		37				
R10	CP to 2Q4		38				
R11	CP to 2TCB		27	4	39		
R12	CP to 2Q3		22	3	28		
R13	CP to 2Q2		21				
R14	CP to 2Q1		25				
R15	CP to 2Q0		26				
R16	CP to 1TC	Propagation delay time (t_{PLH}), C_L = 150 pF	15	5	49		
R17	CP to 1Q3		8	4	38		
R18	CP to 1Q2		12				
R19	CP to 1Q1		13				
R20	CP to 1Q0		14				

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test - Continued.

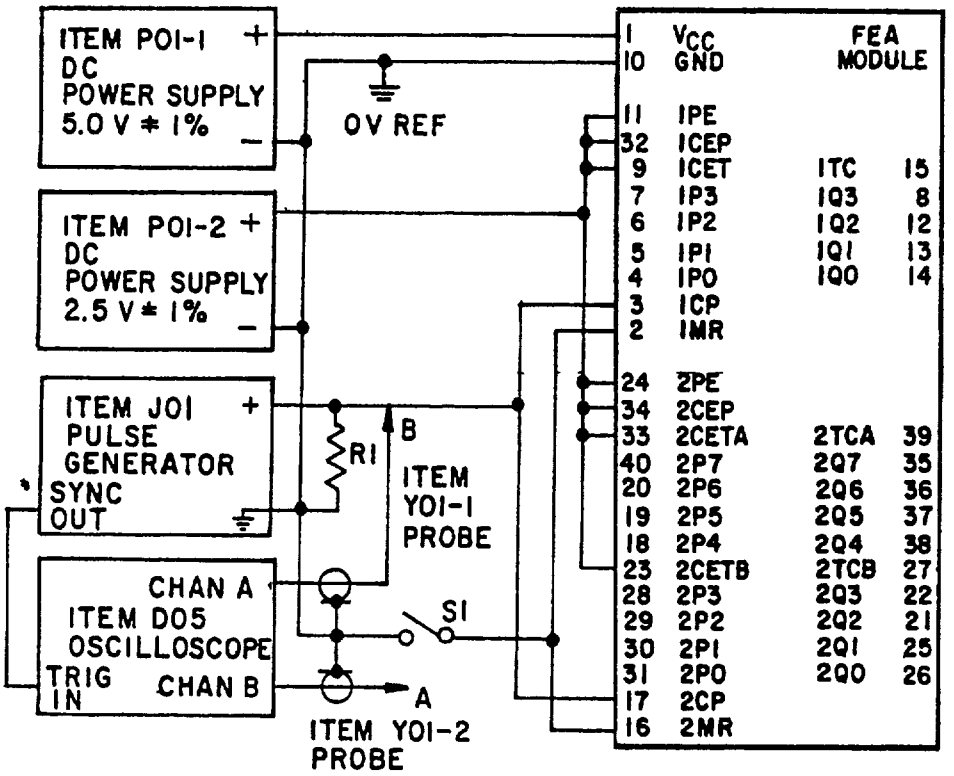
Test condition	Parameter		Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
R21	CP to 2TCA	Propagation delay time (t_{PLH}), C_L = 150 pF	39	5	49	ns	17,18
R22	CP to 2Q7		35	4	38		
R23	CP to 2Q6		36				
R24	CP to 2Q5		37				
R25	CP to 2Q4		38				
R26	CP to 2TCB		27	5	49		
R27	CP to 2Q3		22	4	38		
R28	CP to 2Q2		21				
R29	CP to 2Q1		25				
R30	CP to 2Q0		26				
S1	CP to 1TC	Propagation delay time (t_{PHL}), C_L = 50 pF	15		39		
S2	CP to 1Q3		8		31		
S3	CP to 1Q2		12				
S4	CP to 1Q1		13				
S5	CP to 1Q0		14				
S6	CP to 2TCA		39		39		
S7	CP to 2Q7		35		31		
S8	CP to 2Q6		36				
S9	CP to 2Q5		37				
S10	CP to 2Q4		38				

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TABLE XV. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S11	CP to 2TCB	Propagation delay time (t_{PHL}), C_L = 50 pF	27	4	39	ns	17,18
S12	CP to 2Q3		22		31		
S13	CP to 2Q2		21				
S14	CP to 2Q1		25				
S15	CP to 2Q0		26				
S16	CP to 1TC	Propagation delay time (t_{PHL}), C_L = 150 pF	15	5	49		
S17	CP to 1Q3		8	4	41		
S18	CP to 1Q2		12				
S19	CP to 1Q1		13				
S20	CP to 1Q0		14				
S21	CP to 2TCA		39	5	49		
S22	CP to 2Q7		35	4	41		
S23	CP to 2Q6		36				
S24	CP to 2Q5		37				
S25	CP to 2Q4		38				
S26	CP to 2TCB		27	5	49		
S27	CP to 2Q3		22	4	41		
S28	CP to 2Q2		21				
S29	CP to 2Q1		25				
S30	CP to 2Q0		26				

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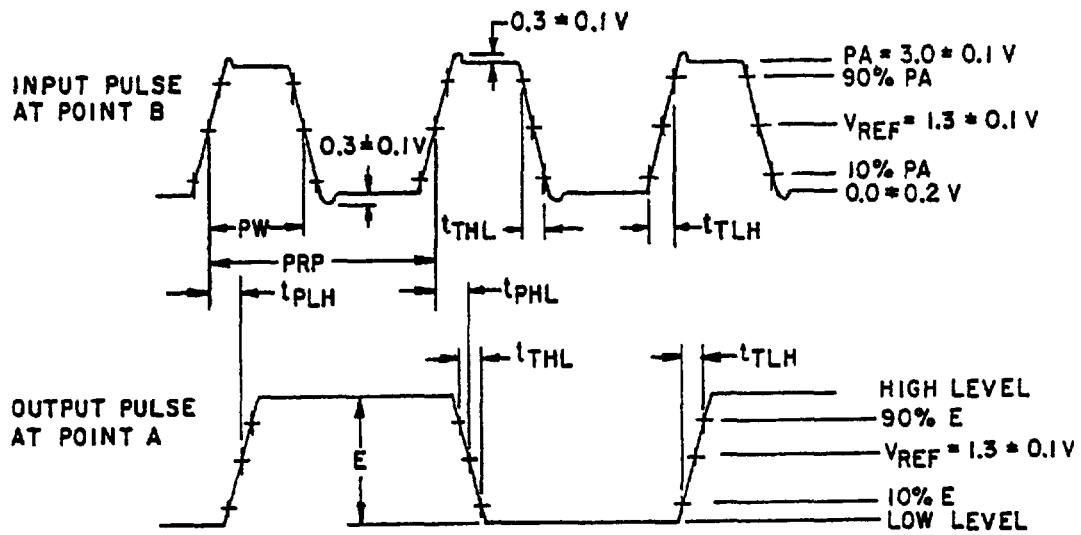


NOTES:

1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. After making all circuit connections for a test condition momentarily close switch S1 for a minimum of 100 ns.
4. All outputs shall be loaded simultaneously per figure 13.

FIGURE 17. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 18. Transition (t_{TLH} and t_{THL}) and propagation delay (t_{PLH} and t_{PHL}) time test circuit waveforms for table XV.

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TABLE XVI. Propagation delay time (t_{PHL}) test.

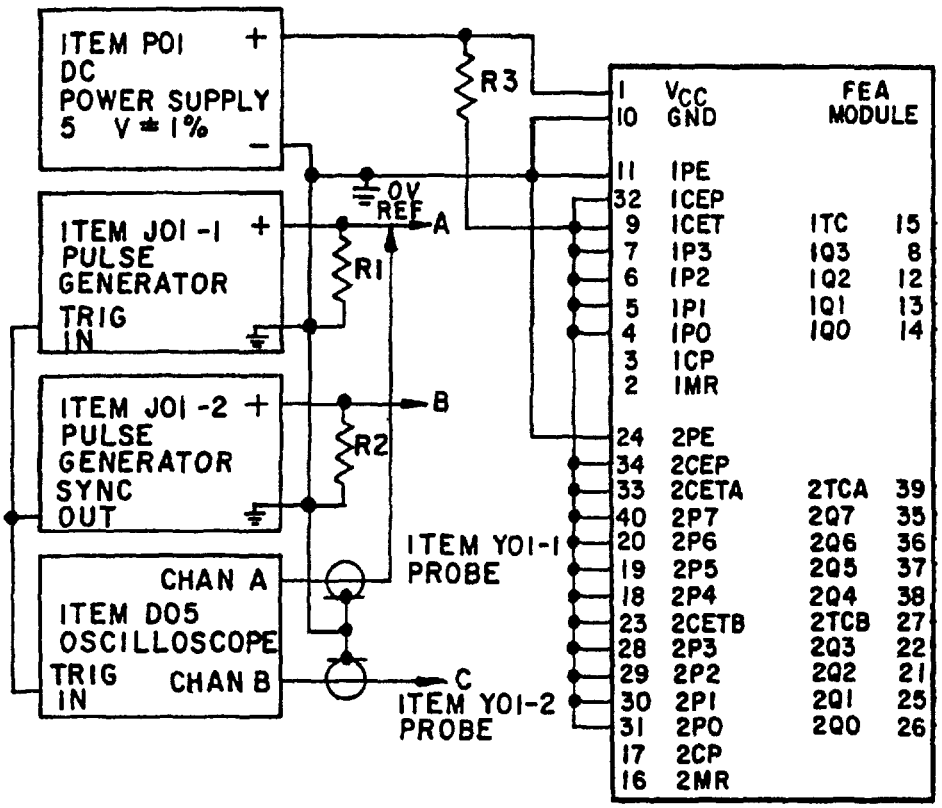
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	MR to 1TC	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3	15	4	42	ns	19,20
T2	MR to 1Q3				8				
T3	MR to 1Q2				12				
T4	MR to 1Q1				13				
T5	MR to 1Q0				14				
T6	MR to 2TCA		16	17	39				
T7	MR to 2Q7				35				
T8	MR to 2Q6				36				
T9	MR to 2Q5				37				
T10	MR to 2Q4				38				
T11	MR to 2TCB				27				
T12	MR to 2Q3				22				
T13	MR to 2Q2				21				
T14	MR to 2Q1				25				
T15	MR to 2Q0				26				
T16	MR to 1TC	Propagation delay time (t_{PHL}), C_L = 150 pF	2	3	15	5	52		
T17	MR to 1Q3				8				
T18	MR to 1Q2				12				
T19	MR to 1Q1				13				
T20	MR to 1Q0				14				

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T21	MR to 2TCA	Propagation delay time (t_{PHL}), C_L = 150 pF	16	17	39	5	52	ns	19,20
T22	MR to 2Q7	↓			35				
T23	MR to 2Q6				36				
T24	MR to 2Q5				37				
T25	MR to 2Q4				38				
T26	MR to 2TCB				27				
T27	MR to 2Q3				22				
T28	MR to 2Q2				21				
T29	MR to 2Q1				25				
T30	MR to 2Q0	↓	↓	↓	26	↓	↓	↓	↓

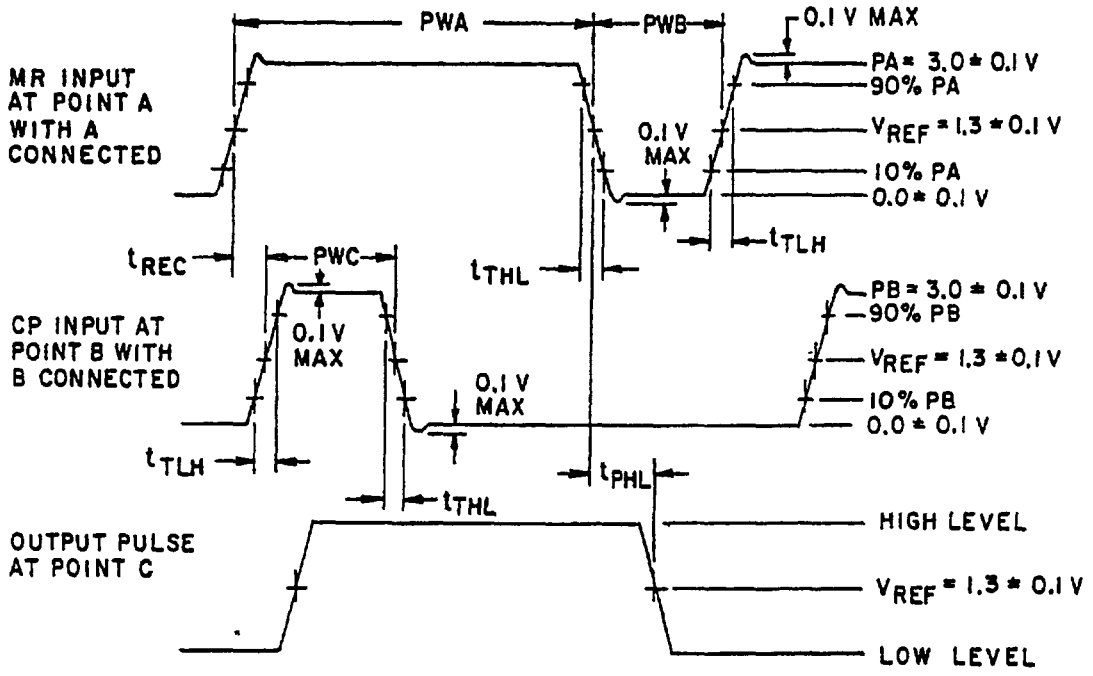
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 1 kΩ ±5%, 1/4 W.
3. All outputs shall be loaded simultaneously per figure 13.

FIGURE 19. Propagation delay time (t_{PHL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 300 ±30 ns.
   PWB = 20 ±1 ns.
   PWC = 100 ±10 ns.
   t_{REC} = 20 ±1 ns.
   $t_{THL} = t_{TLH}$ = 6 ±1 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistors R1 and R2 and input pins; between probes YO1-1 and YO1-2 and input and output pins; and between load circuit to output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

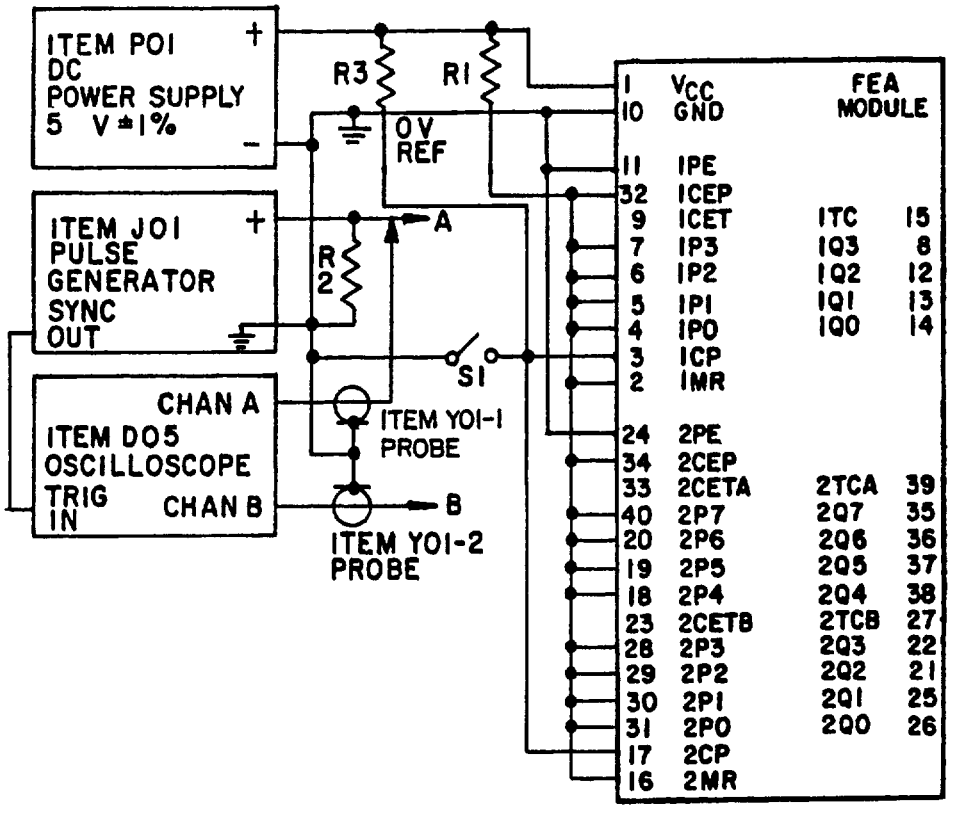
FIGURE 20. Propagation delay time (t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
U1	1CET to 1TC	Propagation delay time (t_{PLH}), C_L = 50 pF	9	15	3	27	ns	21,22
U2	2CETA to 2TCA	Propagation delay time (t_{PLH}), C_L = 50 pF	33	39		27		
U3	2CETB to 2TCB	Propagation delay time (t_{PLH}), C_L = 50 pF	23	27		27		
U4	1CET to 1TC	Propagation delay time (t_{PLH}), C_L = 150 pF	9	15		37		
U5	2CETA to 2TCA	Propagation delay time (t_{PLH}), C_L = 150 pF	33	39		37		
U6	2CETB to 2TCB	Propagation delay time (t_{PLH}), C_L = 150 pF	23	27		37		
V1	1CET to 1TC	Propagation delay time (t_{PHL}), C_L = 50 pF	9	15		27		
V2	2CETA to 2TCA	Propagation delay time (t_{PHL}), C_L = 50 pF	33	39		27		
V3	2CETB to 2TCB	Propagation delay time (t_{PHL}), C_L = 50 pF	23	27		27		
V4	1CET to 1TC	Propagation delay time (t_{PHL}), C_L = 150 pF	9	15		37		
V5	2CETA to 2TCA	Propagation delay time (t_{PHL}), C_L = 150 pF	33	39		37		
V6	2CETB to 2TCB	Propagation delay time (t_{PHL}), C_L = 150 pF	23	27		37		

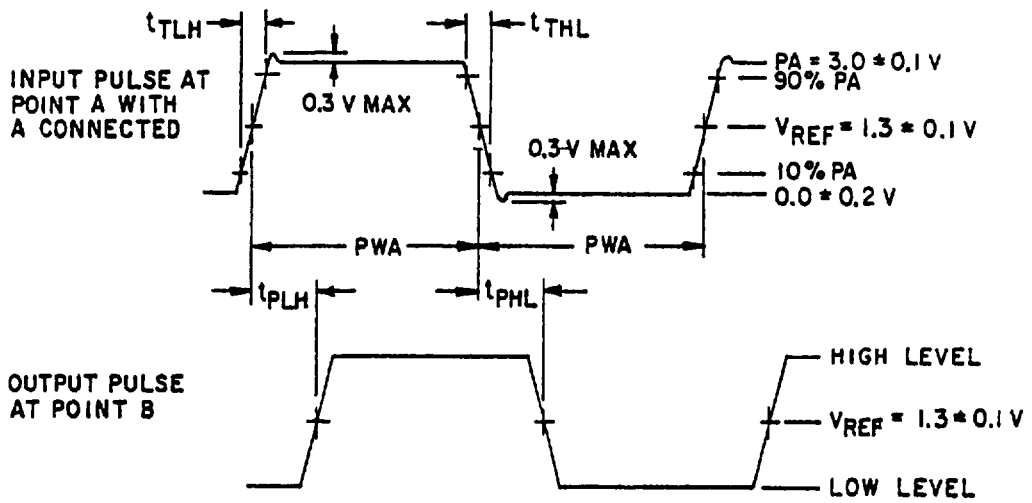
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NOTES:
1. R1 = R3 = 1 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. S1 = SPST switch.
4. Close switch S1 for a minimum of 100 ns, open switch, and then take measurement.
5. All outputs shall be loaded simultaneously per figure 13.

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pin; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

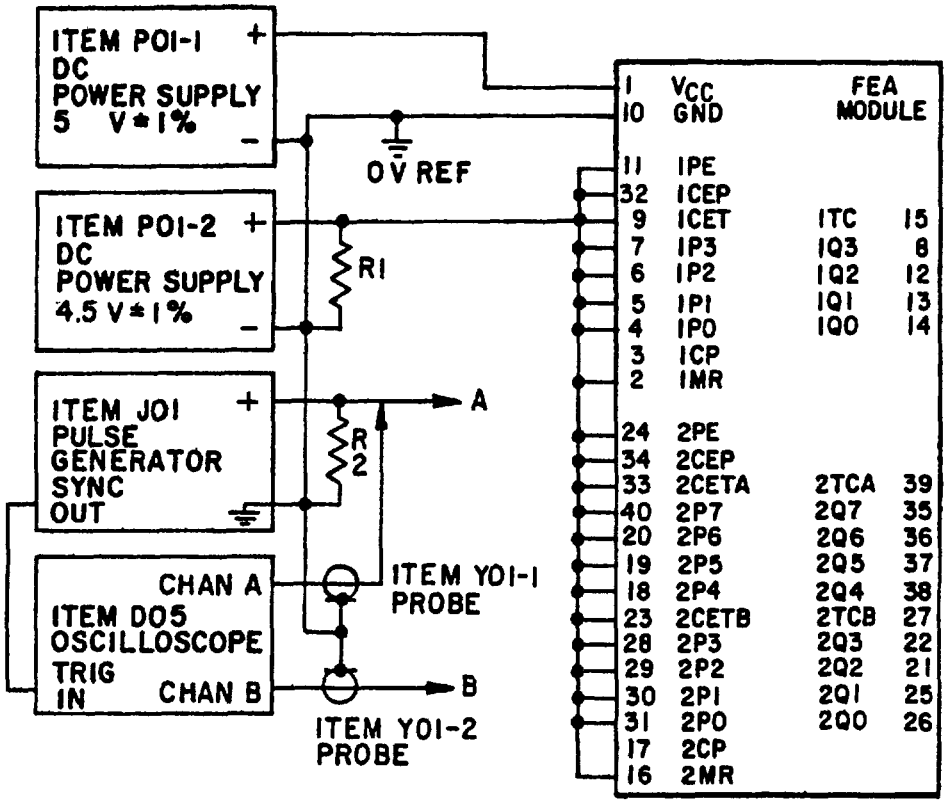
FIGURE 22. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVII.

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* TABLE XVIII. Maximum clock frequency and minimum pulse width test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	1CP	Maximum clock frequency, minimum pulse width, PW1	3	14	48	58	ns	23,24
W2	1CP	Maximum clock frequency, minimum pulse width, PW2	3	14	38	48		
W3	2CP	Maximum clock frequency, minimum pulse width, PW1	17	38	48	58		
W4	2CP	Maximum clock frequency, minimum pulse width, PW2		38	38	48		
W5	2CP	Maximum clock frequency, minimum pulse width, PW1		26	48	58		
W6	2CP	Maximum clock frequency, minimum pulse width, PW2		26	38	48		

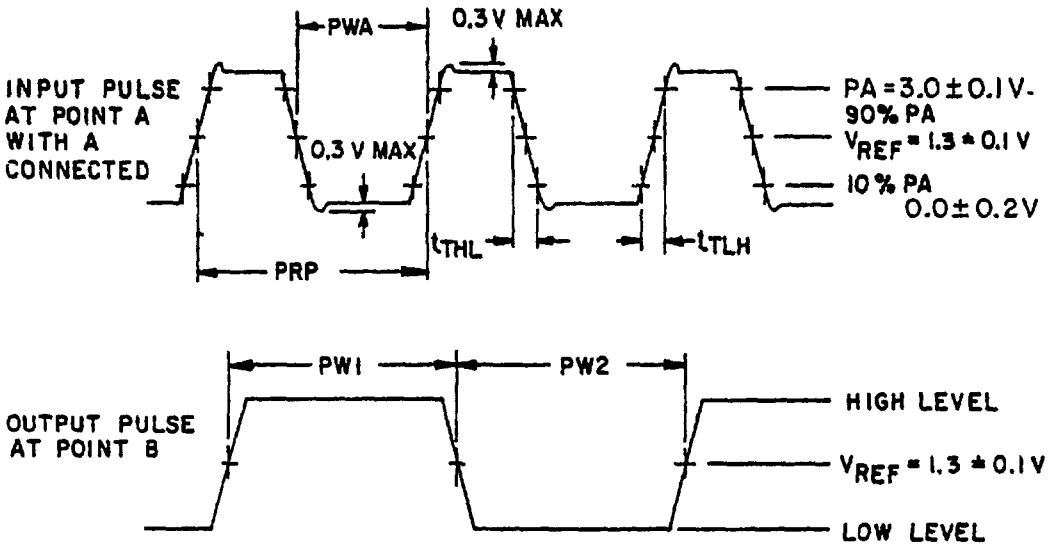
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NOTES:
1. R1 = R2 = 51Ω ± 5%, 1/2 W.
2. Load all outputs per figure 13.

FIGURE 23. Maximum clock frequency and minimum pulse width test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   PWA = 27, +2, -0 ns.
   PRP = 47, +4, -0 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pin; between probes YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.
5. Output waveform shall be stable, indicating proper switching.

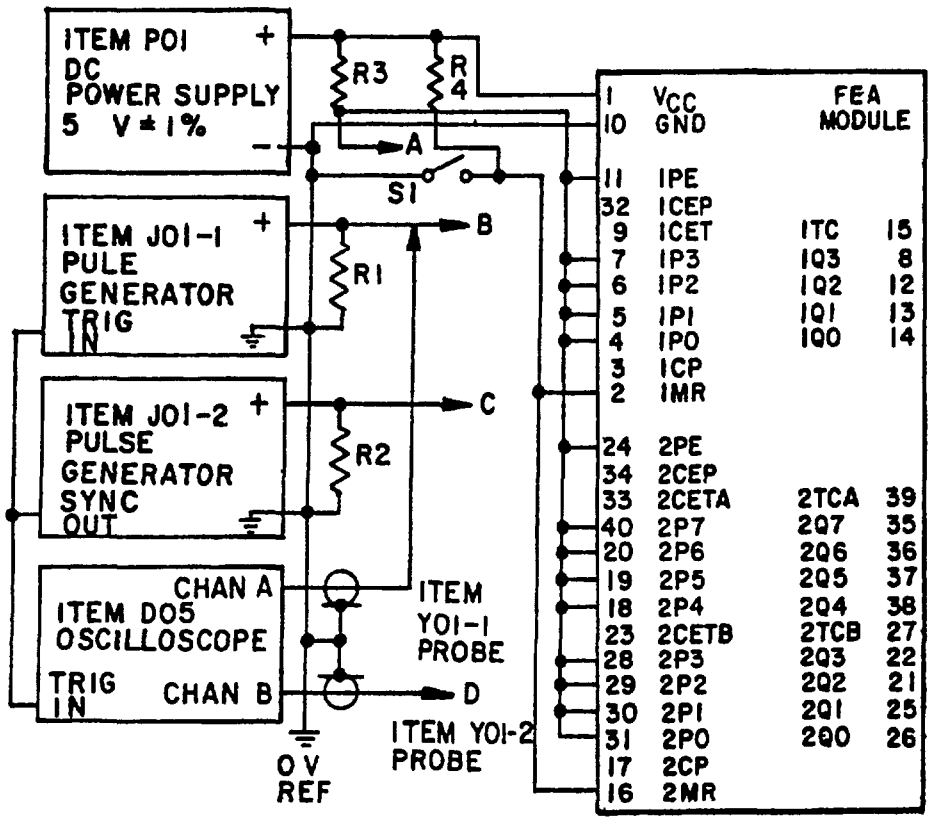
FIGURE 24. Maximum clock frequency and minimum pulse width test circuit waveforms for table XVIII.

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TABLE XIX. Setup (t_S) and hold (t_H) time check test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1CET	Setup time (t_S) check measuring t_C	32	9	3	8	3,100	3,300	ns	25,26
X2	2CETA	↓	34	33	17	35	↓	↓	↓	↓
X3	2CETB		34	23	17	22				
X4	1CEP		9	32	3	8				
X5	2CEP		33	34	17	35				
X6	2CEP		23	34	17	22				
Y1	1CET	Hold time (t_H) check measuring t_C	32	9	3	8				
Y2	2CETA	↓	34	33	17	35				
Y3	2CETB		34	23	17	22				
Y4	1CEP		9	32	3	8				
Y5	2CEP		33	34	17	35				
Y6	2CEP		23	34	17	22				

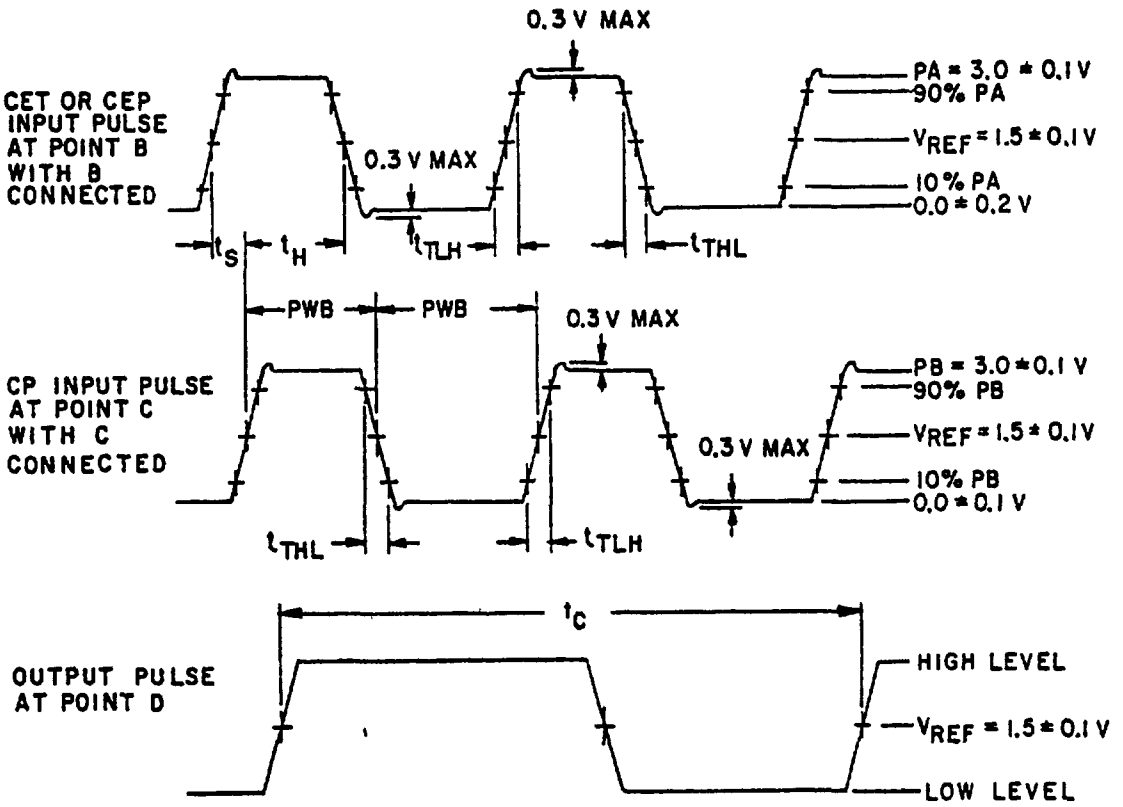
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = R4 = 1 kΩ ±5%, 1/4 W.
3. S1 = SPST switch.
4. After all circuit connections are made, close switch S1 for a minimum of 100 ns, then open S1 and make measurement.
5. Load all outputs per figure 13.

FIGURE 25. Setup (t_S) and hold (t_H) time test circuit for table XIX.

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NOTES:
1. Input pulse parameters for test conditions X1-X6:
   t_S = 35 ±3 ns.
   t_H = 30 ±3 ns.
   PWB = 100 ±6 ns.
   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
2. Input pulse parameters for test conditions Y1-Y6:
   t_S = 40 ±4 ns.
   t_H = 10 ±1 ns.
   PWB = 100 ±6 ns.
   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
3. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
4. The lead length of electrical conductors between resistors R1 and R2 and input pins; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to ground.

FIGURE 26. Setup (t_S) and hold (t_H) time check test circuit waveforms for table XIX.

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TABLE XX. Setup and hold time (PE and P) check test.

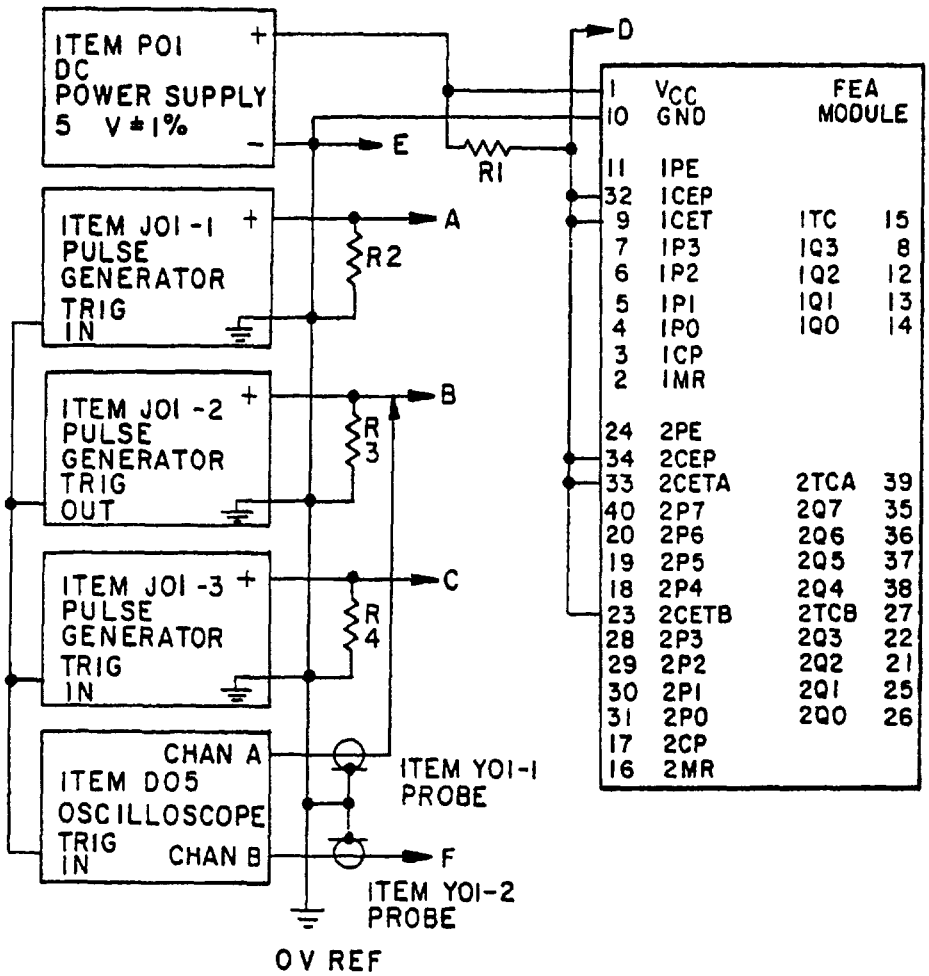
Test condition		Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Z1	1PE	Setup and hold time check, t_{PLH}	11	3	2	4,5,6,7	Open	8	3	29	ns	27,28
Z2	2PE	Setup and hold time check, t_{PLH}	24	17	16	18,19, 20,40	Open	35		29		27,28
Z3	3PE	Setup and hold time check, t_{PLH}	24	17	16	28,29, 30,31	Open	22		29		27,28
Z4	1P0	Setup and hold time check, low level, t_{PHL}	Open	4	3	2,5,6,7	11	14		31		27,29
Z5	1P1			5		2,4,6,7		13				
Z6	1P2			6		2,4,5,7		12				
Z7	1P3			7		2,4,5,6		8				
Z8	2P7			40	17	16,18, 19,20	24	35				
Z9	2P6			20		16,18, 19,40		36				
Z10	2P5			19		16,18, 20,40		37				
Z11	2P4			18		16,19, 20,40		38				
Z12	2P3			28		16,29, 30,31		22				
Z13	2P2			29		16,28, 30,31		21				
Z14	2P1			30		16,28, 29,31		25				
Z15	2P0			31		16,28, 29,30		26				

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TABLE XX. Setup and hold time (PE and P) check test - Continued.

Test condition		Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Z16	1P3	Setup and hold time check, high level, t_{PLH}	7	3	Open	2,4,5,6	11	8	3	28	ns	27,30
Z17	1P2		6			2,4,5,7		12				
Z18	1P1		5			2,4,6,7		13				
Z19	1P0		4			2,5,6,7		14				
Z20	2P7		40	17		16,18, 19,20	24	35				
Z21	2P6		20			16,18, 19,40		36				
Z22	2P5		19			16,18, 20,40		37				
Z23	2P4		18			16,19, 20,40		38				
Z24	2P3		28			16,29, 30,31		22				
Z25	2P2		29			16,28, 30,31		21				
Z26	2P1		30			16,28, 29,31		25				
Z27	2P0		31			16,28, 29,30		26				

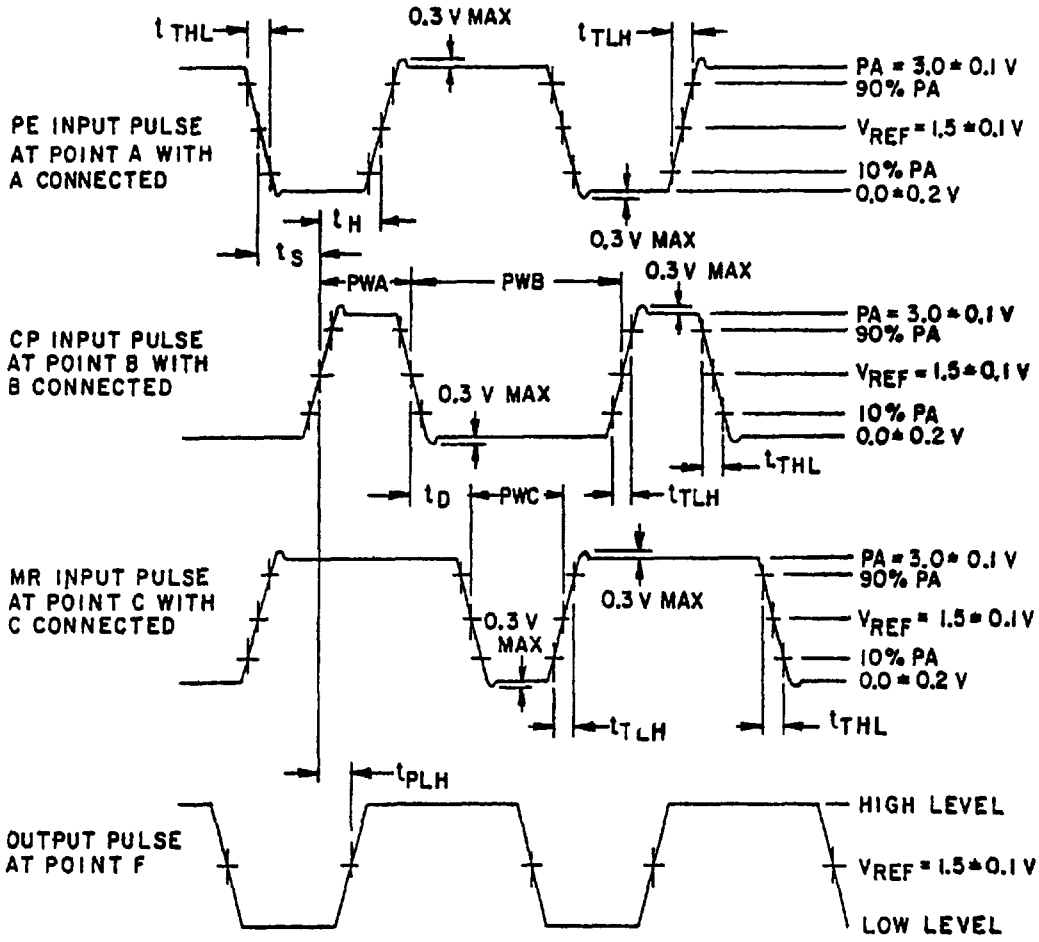
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NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = R3 = R4 = 51Ω ±5%, 1/2 W.
3. Load all outputs per figure 13.

FIGURE 27. Setup and hold time (PE and P) check test circuit for table XX.

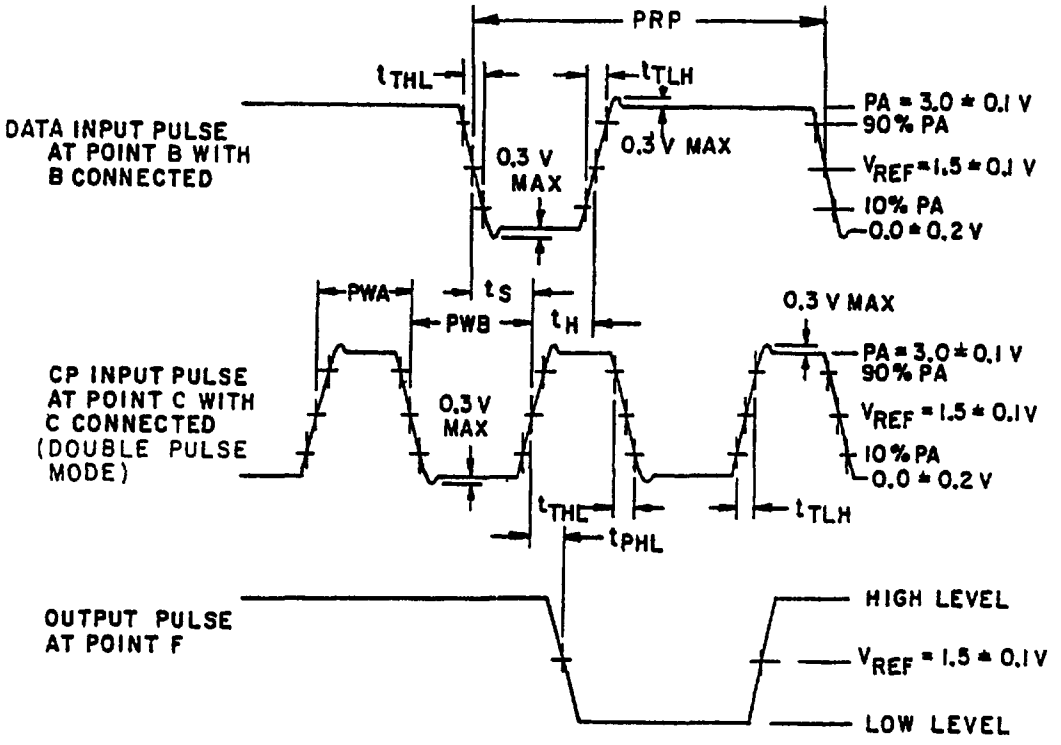
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NOTES:
1. Input pulse parameters for test conditions Z1-Z3:
   PWA = 100 ±10 ns.
   PWB = 200 ±20 ns.
   PWC = 100 ±10 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   $t_S = 25 \pm 1$ ns.
   $t_H = 10 \begin{smallmatrix} +1 \\ -0 \end{smallmatrix}$ ns
   $t_D = 50 \pm 5$ ns.
2. Before making the measurements , verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistors R2, R3, and R4 and input pins; between probes YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 28. Setup and hold time (PE) check test circuit waveforms for table XX.

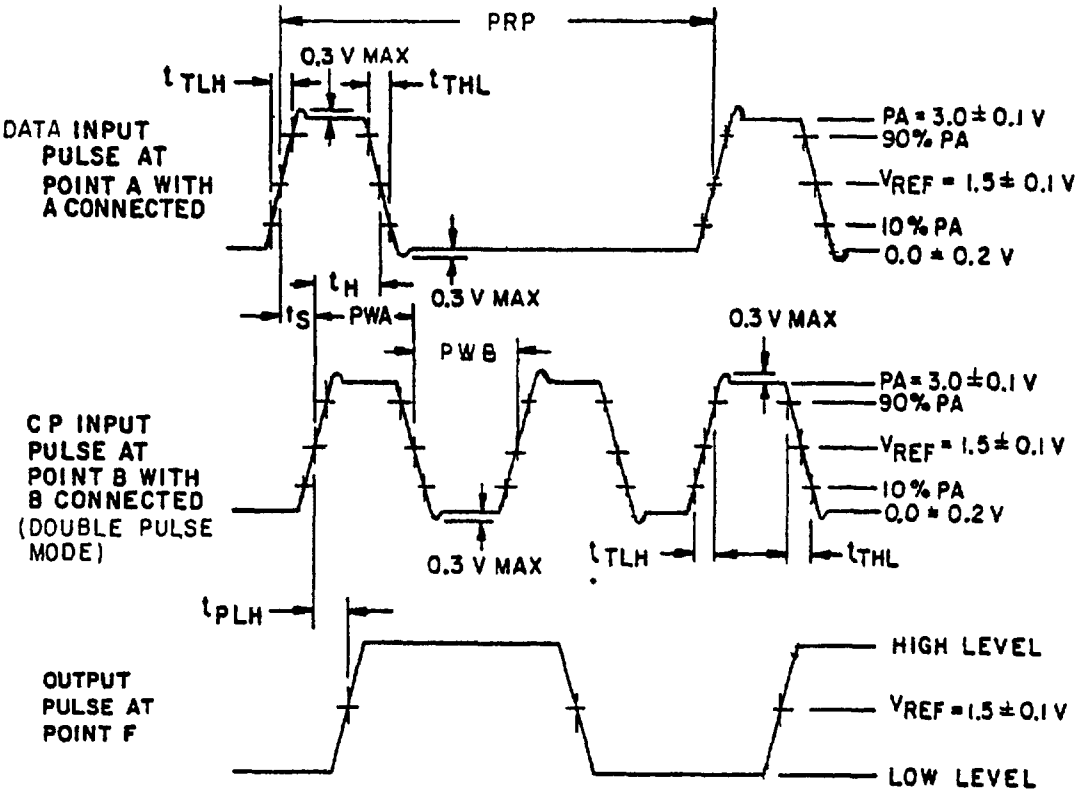
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NOTES:
1. Input pulse parameters for test conditions Z4-Z15:
   PWA = 100 ±10 ns.
   PWB = 100 ±10 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   $t_S = 20 \begin{smallmatrix}+2 \text{ ns} \\ -0 \text{ ns}\end{smallmatrix}$
   $t_H = 3 \begin{smallmatrix}+1 \text{ ns} \\ -0 \text{ ns}\end{smallmatrix}$
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistors R2, R3, and R4 and input pins; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.
5. Item J01-3, double pulse mode.

FIGURE 29. Setup and hold time (P, low level) check test circuit waveforms for table XX.

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NOTES:
1. Input pulse parameters for test conditions Z16-Z27:
   PWA = 100 ±10 ns.
   PWB = 100 ±10 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   $t_S = 20 \begin{smallmatrix} +2 \text{ ns} \\ -0 \text{ ns} \end{smallmatrix}$
   $t_H = 3 \begin{smallmatrix} +1 \text{ ns} \\ -0 \text{ ns} \end{smallmatrix}$
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistors R2, R3, and R4 and input pins; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.
5. Item J01-2, double pulse mode.

FIGURE 30. Setup and hold time (P, high level) check test circuit waveforms for table XX.

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TABLE XXI. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA1	1TC	Output crosstalk, low level	23,33	9	40,31, 30,29, 28,20, 19,18, 7,6,5,4	3,17	15	0	400	mV	31,32
AA2	1Q3		9,23,33	7	40,31, 30,29, 28,20, 19,18, 6,5,4		8				
AA3	1Q2			6	40,31, 30,29, 28,20, 19,18, 7,5,4		12				
AA4	1Q1			5	40,31, 30,29, 28,20, 19,18, 7,6,4		13				
AA5	1Q0			4	40,31, 30,29, 28,20, 19,18, 7,6,5		14				
AA6	2TCA		9,23	33	40,31, 30,29, 28,20, 19,18, 7,6,5,4		39				
AA7	2Q7		9,23,33	40	31,30, 29,28, 20,19, 18,7,6, 5,4		35				
AA8	2Q6		9,23,33	20	40,31, 30,29, 28,19, 18,7,6, 5,4		36				

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TABLE XXI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
AA9	2Q5	Output crosstalk, low level	9,23,33	19	40,31, 30,29, 28,20, 18,7,6, 5,4	3,17	37	0	400	mV	31,32
AA10	2Q4		9,23,33	18	40,31, 30,29, 28,20, 19,7,6, 5,4		38				
AA11	2TCB		9,33	23	40,31, 30,29, 28,20, 19,18, 7,6,5,4		27				
AA12	2Q3		9,23,33	28	40,31, 30,29, 20,19, 18,7,6, 5,4		22				
AA13	2Q2			29	40,31, 30,28, 20,19, 18,7,6, 5,4		21				
AA14	2Q1			30	40,31, 29,28, 20,19, 18,7,6, 5,4		25				
AA15	2Q0			31	40,30, 29,28, 20,19, 18,7,6, 5,4		26				

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TABLE XXI. Output crosstalk (low-level and high-level) test - Continued.

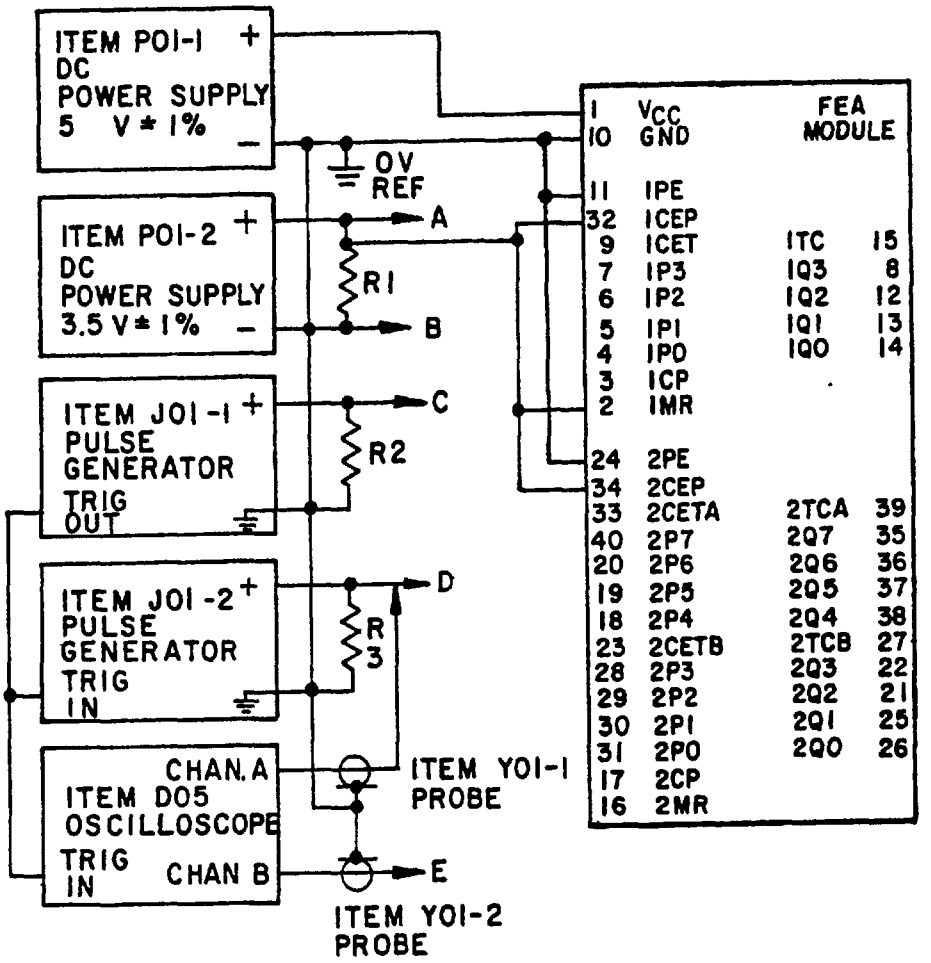
Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	1Q3	Output crosstalk, high level	7,9,23, 33	Open	40,31, 30,29, 28,20, 19,18, 6,5,4	3,17	8	0	400	mV	31,32
BB2	1Q2		6,9,23, 33		40,31, 30,29, 28,20, 19,18, 7,5,4		12				
BB3	1Q1		5,9,23, 33		40,31, 30,29, 28,20, 19,18, 7,6,4		13				
BB4	1Q0		4,9,23, 33		40,31, 30,29, 28,20, 19,18, 7,6,5		14				
BB5	2Q7		9,23, 33,40		31,30, 29,28, 20,19, 18,7, 6,5,4		35				
BB6	2Q6		9,20, 23,33		40,31, 30,29, 28,19, 18,7,6, 5,4		36				
BB7	2Q5		9,19, 23,33		40,31, 30,29, 28,20, 18,7,6, 5,4		37				

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TABLE XXI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB8	2Q4	Output crosstalk, high level	9,18, 23,33	Open	40,31, 30,29, 28,20, 19,7,6, 5,4	3,17	38	0	400	mV	31,32
BB9	2Q3		9,23, 28,33		40,31, 30,29, 20,19, 18,7,6, 5,4		22				
BB10	2Q2		9,23, 29,33		40,31, 30,28, 20,19, 18,7,6, 5,4		21				
BB11	2Q1		9,23, 30,33		40,31, 29,28, 20,19, 18,7,6, 5,4		25				
BB12	2Q0		9,23, 31,33		40,30, 29,28, 20,19, 18,7,6, 5,4		26				

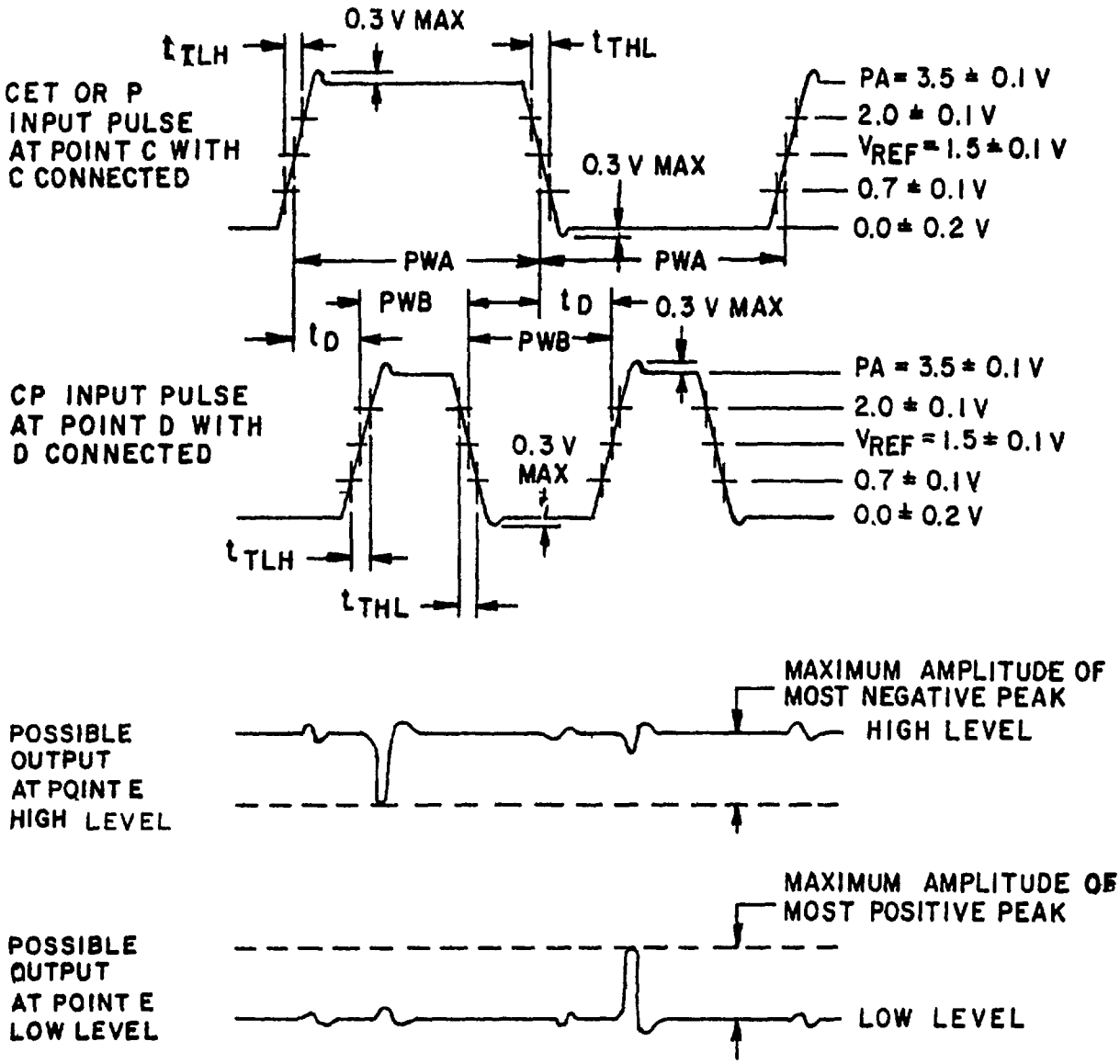
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NOTES:
1. R1 = R2 = R3 = 51Ω ±5%, 1/4 W.
2. Load all outputs with load as specified on figure 13 with the exception that C_L is 50 ±5 pF.

FIGURE 31. Output crosstalk (low-level and high-level) test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   t_D = 25 ±3 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistors R2 and R3 and input pins; between probes YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).

FIGURE 32. Output crosstalk (low-level and high-level) test circuit waveforms for table XXI.

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
CC	Power supply transient amplitude	0	400	mV	33,34

TABLE XXIII. Power supply filter capacitance test.

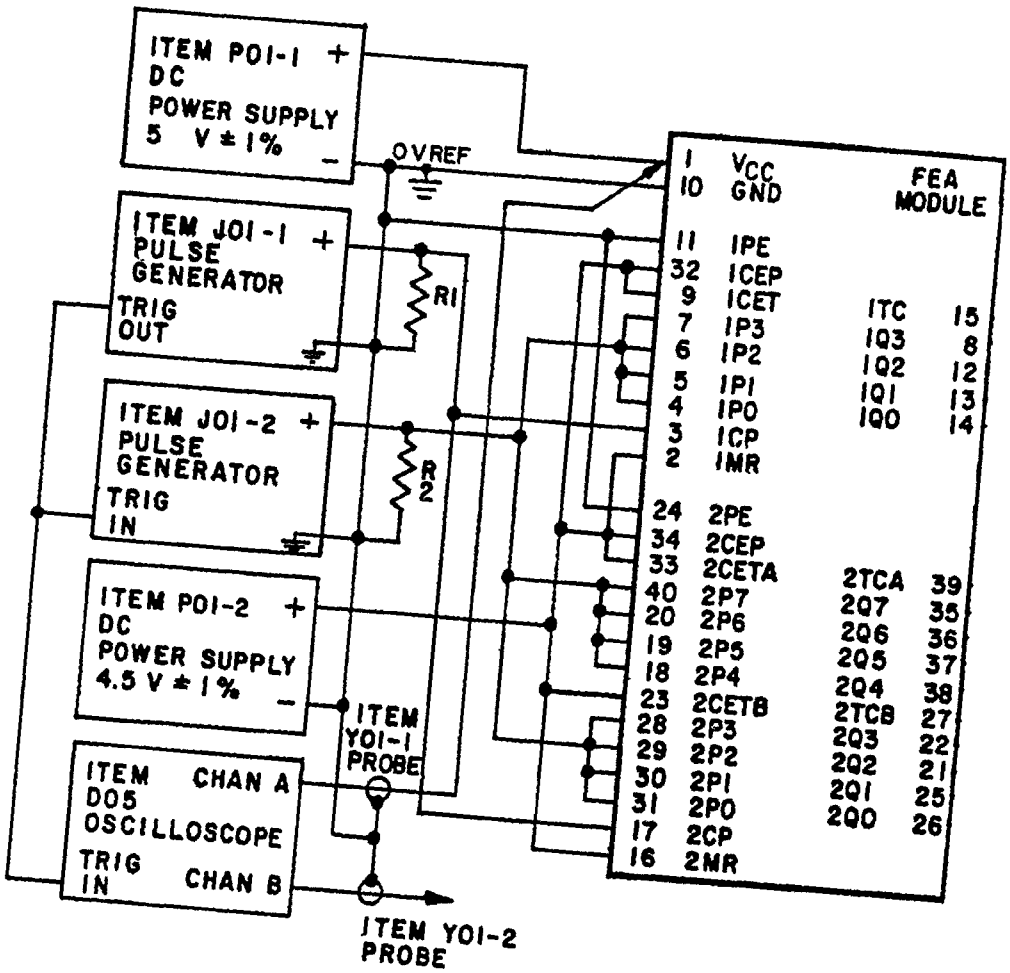
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
DD	Power supply filter capacitance	Open	1	1.6	3.0	µF	35

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* TABLE XXIV. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25 ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
EE1	1PE Input capacitance	11	Open	2	25	pF	35
EE2	1CEP	32					
EE3	1CET	9					
EE4	1P3	7					
EE5	1P2	6					
EE6	1P1	5					
EE7	1PO	4					
EE8	1CP	3					
EE9	1MR	2					
EE10	2PE	24		4	40		
EE11	2CEP	34		4	40		
EE12	2CETA	33		2	25		
EE13	2P7	40					
EE14	2P6	20					
EE15	2P5	19					
EE16	2P4	18					
EE17	2CETB	23					
EE18	2P3	28					
EE19	2P2	29					
EE20	2P1	30					
EE21	2PO	31					
EE22	2CP	17		4	40		
EE23	2MR	[illegible]		4	40		

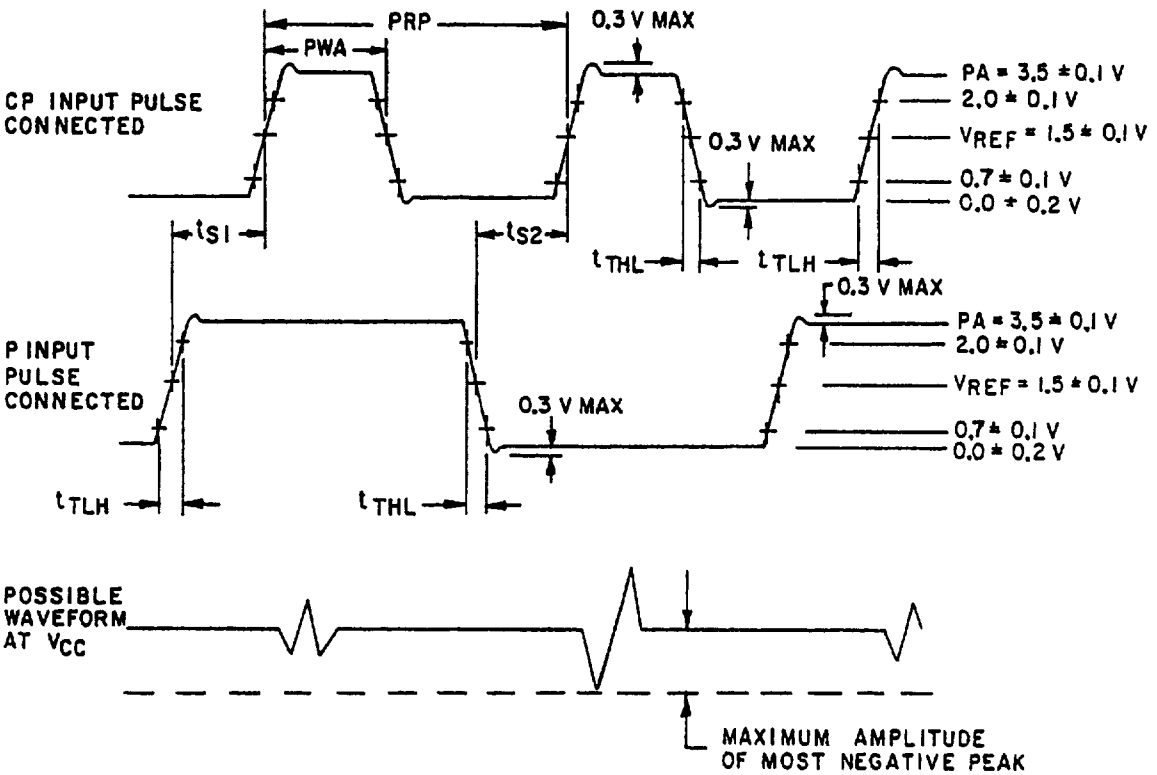
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 33. Power supply transient amplitude test circuit for table XXII.

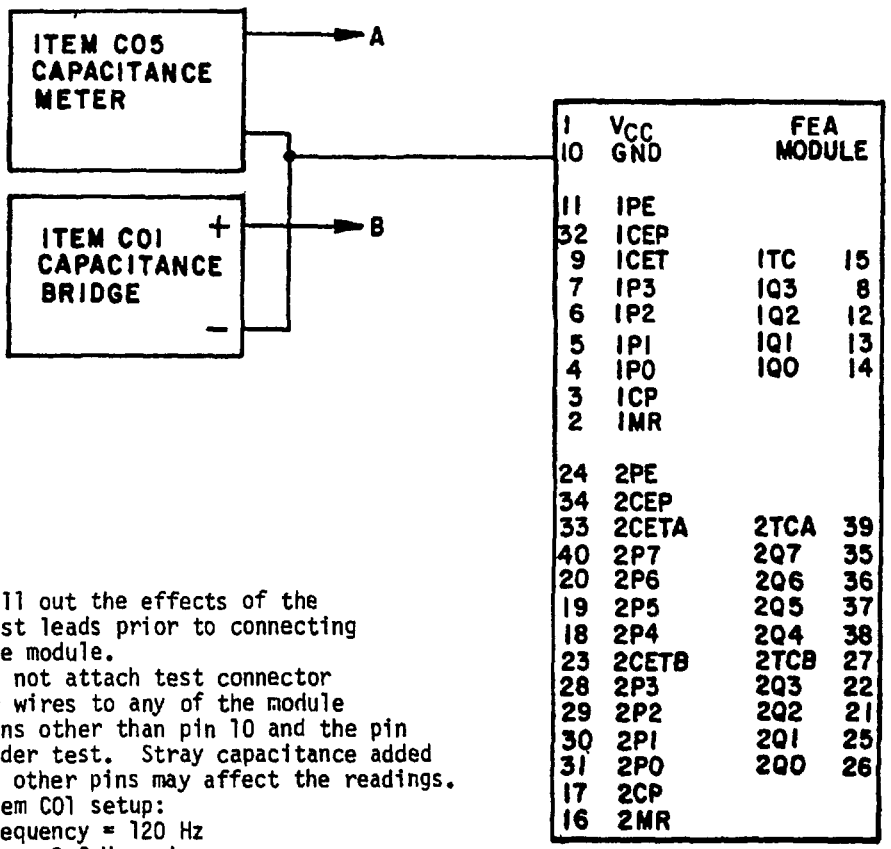
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NOTE: Input pulse parameters:
PWA = 250 ±25 ns.
PRP = 500 ±50 ns.
$t_{S1} = t_{S2}$ = 50 ±5 ns.
$t_{TLH} = t_{THL}$ = Vary t_{TLH} from 5 to 25 ns and t_{THL} from 3 to 10 ns and record maximum value of transient amplitude.

FIGURE 34. Power supply transient amplitude test circuit waveforms for table XXII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz
   V_{pk} = 0.3 V maximum.

FIGURE 35. Power supply filter and input capacitance test circuit for table XXIII and XXIV.

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